

RAG AND EPR

Retrieval-Augmented Generation, Evidence
Portfolio Retrieval, and Research Automation

Architectures, Failure Modes, Evaluation Methods, and a Research Programme

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RAG • EVIDENCE RETRIEVAL • SCIENTIFIC REASONING • EPR

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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboai. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test and popularise our latest ideas and technologies.

Abstract

Retrieval-Augmented Generation has become a principal architecture for connecting large language models to private, current, or domain-specific knowledge. The term now covers systems that differ substantially in how they index sources, interpret queries, retrieve candidates, rerank evidence, construct context, invoke models, verify claims, and measure quality. This book provides an engineering account of that landscape as it stands in August 2026. It explains sparse, dense, hybrid, and late-interaction retrieval; chunking and hierarchical indexing; reranking, query expansion, multi-hop and graph retrieval; adaptive and agentic control; long-context trade-offs; citation and faithfulness mechanisms; security and operational concerns; and the evaluation methods required to distinguish robust implementations from inadequately controlled prototypes.

The second part defines Evidence Portfolio Retrieval, or EPR, as a separate research programme motivated by the automation of scientific work. Conventional RAG usually ranks documents individually by estimated relevance to a query. EPR asks which bounded set of evidence units, considered collectively, is most useful for a downstream epistemic decision. A portfolio for novelty assessment may require a close holistic precedent, component precedents, a structural analogue, a boundary case, contradictory evidence, and an explicit account of what remains uncovered. A portfolio for peer review may require support, replication, methodological criticism, negative results, and alternative explanations. This changes the unit of optimization from an independently scored document to a budget-constrained set whose collective coverage, independence, provenance, and sufficiency can be inspected.

EPR is treated as a testable research hypothesis rather than as an established replacement for RAG. Modern RAG already includes diversification, late interaction, multi-hop retrieval, subgraph selection, iterative search, evidence reranking, and scientific synthesis agents. EPR is scientifically relevant only if explicit epistemic-role and evidence-sufficiency objectives improve downstream decisions at the same context, cost, and latency budget over competitive baselines such as hybrid retrieval, CoBERTv2-style late interaction, cross-encoder reranking, maximal marginal relevance, learned diversification, agentic query decomposition, and LLM-based evidence selection. The resulting programme is both constructive and comparative: it specifies how EPR can be built and how the hypothesis can be rejected if its additional machinery does not improve measurable outcomes.

Preface: Scope, Motivation, and Positioning

RAG is frequently summarized through three stages: retrieve, augment, and generate. This abstraction identifies the category but omits many engineering choices that determine accuracy, reproducibility, security, cost, and scientific defensibility. Two applications described as RAG may share few implementation decisions. One may use BM25 over carefully authored sections, a cross-encoder, a deterministic extractive reader, and source-level access controls. Another may embed fixed-size chunks, select five nearest neighbours, concatenate them into a prompt, and request an answer from a general model. Their error profiles and operational guarantees are not comparable.

The broad use of the term also creates an attribution problem. Improvements are sometimes assigned to RAG although they arise from better corpora, document parsing, query rewriting, larger context windows, stronger language models, rerankers, additional tool calls, or evaluation leakage. Conversely, a disappointing result may be attributed to retrieval as a class when the immediate cause is incomplete evidence, unsuitable chunks, poor calibration, or a generator that does not use the supplied context. A technical account must therefore decompose the pipeline and preserve causal attribution.

Scientific automation requires capabilities beyond fluent response generation. A research assistant must locate evidence dispersed across papers, distinguish claims from interpretations, preserve source spans, handle incompatible results, reason about versions and dates, avoid confusing absence of evidence with negative evidence, and expose the limits of corpus coverage. In novelty assessment, semantic distance cannot be treated as proof of novelty. In peer review, repeated retrieval from one intellectual lineage should not be presented as independent consensus. In scientific discovery, an unsearched region of the literature should not be reported as a research gap. These requirements concern retrieval, generation, and epistemic governance simultaneously.

The Evidence Portfolio Retrieval programme is motivated by a recurring limitation in complex research search: the highest-ranked documents are not always the most useful set of documents. The contribution of an item depends on what has already been selected. A moderately similar paper that provides a missing negative result, mechanism, methodological limitation, or boundary condition may be more useful for a later decision than another highly similar account of an already covered claim. EPR formalizes this set-level information need without requiring a new semantic geometry or a replacement for modern retrieval infrastructure.

This volume has two purposes. It is a tutorial and design reference for modern RAG, written for readers who need to build and evaluate complete systems. It also positions EPR within a larger programme of executable and automatable science. EPR does not replace candidate retrieval, language models, symbolic reasoning, or expert judgement. It occupies the boundary between retrieval and reasoning, where a system determines which evidence enters a fixed context budget and how that evidence is organized for the decision that follows.

The exposition follows an engineering sequence. Formal objectives appear only after the task, architecture, and evaluation requirements have been established. Equations can specify budgets, constraints, and optimization criteria, but they cannot compensate for an

inadequate definition of relevance, an unrepresentative corpus, or a benchmark whose labels encode only one acceptable research path. The operational questions therefore come first: which evidence unit is indexed, which task is supported, which failure is measurable, which baseline is credible, and which empirical result would justify continuation or termination of the programme.

Contents

Chapter 1. Retrieval-Augmented Generation as a Systems Architecture	1
Chapter 2. Sparse, Dense, Hybrid, and Late-Interaction Retrieval	4
Chapter 3. Context Construction, Grounding, and Generation	7
Chapter 4. Advanced Retrieval-Augmented Architectures	10
Chapter 5. Failure Modes Across the RAG Pipeline	13
Chapter 6. Evaluation Methodology and Benchmark Design	17
Chapter 7. Limits of Independent Top-k Ranking in Scientific Tasks	20
Chapter 8. Evidence Portfolio Retrieval: Definition and Scope	24
Chapter 9. Architectures and Design Patterns for Evidence Portfolios	28
Chapter 10. Evidence Portfolio Retrieval in Scientific Automation	32
Chapter 11. Formal Objectives and Optimization Methods	37
Chapter 12. Comparative Evaluation Against Modern Retrieval Baselines	41
Chapter 13. Governance, Security, Provenance, and Accountability	46
Chapter 14. Research Agenda and Technology Development	49
Chapter 15. Conclusions: Retrieval as Evidence Selection for Scientific Decisions	54

Chapter 1. Retrieval-Augmented Generation as a Systems Architecture

Retrieval-Augmented Generation began as a precise model class. Lewis and colleagues combined a sequence-to-sequence generator with a differentiable retriever over an external corpus, treating documents as latent variables and showing that parametric and non-parametric memory could be trained jointly for knowledge-intensive generation [1]. In contemporary practice, the name has broadened. It now describes almost any architecture in which a model receives information selected from a source outside its weights. This broad definition is useful, provided that the system is decomposed into explicit contracts rather than treated as a monolith.

A useful engineering definition is that RAG is a controlled information-flow system. A corpus is acquired and normalized. Documents are segmented into retrieval units. Those units are represented in one or more indexes. A query is interpreted and possibly rewritten. A candidate generator retrieves a pool. A reranker or selector reduces the pool. A context constructor orders, compresses, labels, and packages the selected evidence. A generator produces an answer. A verifier may check claims, citations, constraints, or confidence. The answer and its trace are then exposed to a user or another agent. Each boundary can introduce or remove error.

RAG is a pipeline, not a single technique

Every boundary can add or remove evidence

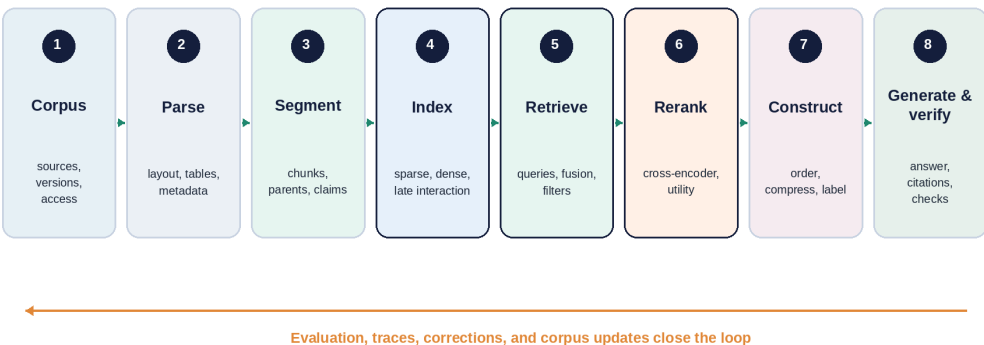


Figure 1. The operational RAG stack. Performance and provenance depend on the contracts between stages, not only on the retriever or generator.

The basic architectural motivation is the separation of parametric and external knowledge. A language model's weights encode broad regularities and facts acquired

during training, but updates are expensive, provenance is largely unavailable, and access control cannot be expressed at the level of individual remembered passages. An external corpus can be revised, versioned, filtered, and audited. Retrieval makes current or private material available at inference time without requiring every change to be absorbed through fine-tuning. This does not mean that the model becomes a transparent reasoner. It means that part of the evidence used for an answer can be made explicit.

Early RAG developed alongside dense retrieval. Dense Passage Retrieval showed that a dual encoder trained on question-passage pairs could outperform a strong BM25 baseline on several open-domain question-answering datasets [2]. REALM integrated retrieval into language-model pretraining [3]. Fusion-in-Decoder encoded retrieved passages separately and fused them in the decoder, making multi-passage aggregation effective while avoiding a single quadratic encoder over the concatenated context [4]. RETRO demonstrated a different scale regime in which a language model retrieves neighbours from a database containing trillions of tokens and conditions generation on them [5]. These systems established several enduring patterns: retrieval can be trained or frozen, evidence can be consumed before or during generation, and the retrieved unit may be a passage, neighbour chunk, structured fact, image, table, or graph fragment.

The contemporary engineering stack is less unified. A production system may use BM25 for exact terminology, a dense encoder for semantic recall, reciprocal-rank fusion to combine them, a cross-encoder to rerank the top hundred candidates, a language model to decompose the question, and a second model to verify the final citations. The retriever may search a vector database, a Lucene index, a knowledge graph, a SQL store, a web engine, or several of these. RAG is consequently better understood as an architectural family than as a single algorithm.

This family has four different objectives that should not be collapsed. The first is knowledge access: retrieving information the model does not reliably contain. The second is grounding: constraining an answer to externally visible evidence. The third is adaptation: changing the effective knowledge base without retraining the generator. The fourth is computation: using retrieval as an intermediate operation in multi-step reasoning or agentic search. An implementation optimized for one objective may be poor at another. For example, a system that retrieves one exact policy clause can be excellent for access and grounding but inadequate for comparing conflicting policies across time. A system that recursively explores a literature graph may support synthesis but be wasteful for exact lookup.

RAG does not by itself guarantee factuality. Relevant context can be absent, misleading, outdated, poisoned, internally inconsistent, or ignored by the generator. The model can cite a source that does not entail the claim, merge incompatible evidence, or prefer its parametric memory when retrieved text disagrees. Studies of context-memory conflict and citation faithfulness show that providing documents is not equivalent to using them correctly [42–44]. The engineering claim is therefore conditional: RAG provides a mechanism for grounded generation, while grounding quality depends on the corpus, retrieval, context construction, generation policy, and verification layers.

The corpus is part of the model. This point is frequently underestimated. The behaviour of a RAG system is determined not only by the neural components but by which sources are admitted, how they are parsed, which versions coexist, how metadata

is normalized, and whether retractions, corrections, duplicates, and access policies are represented. In scientific use, the corpus may contain preprints, accepted manuscripts, final versions, supplementary material, datasets, review reports, patents, protocols, and laboratory records. Treating all of these as interchangeable text fragments destroys distinctions that matter to later reasoning.

The query is likewise an object of interpretation. A user's natural-language request may contain multiple tasks, ambiguous entities, implicit time constraints, and assumptions that should not be carried into retrieval. The literal words may be poor search terms. HyDE showed that a generated hypothetical document can bridge the representational mismatch between short questions and relevant documents in zero-shot dense retrieval [11]. Query expansion, decomposition, entity resolution, and intent classification can improve recall, but they can also introduce unsupported concepts. A robust system records the original query, each transformation, and the retrieval result produced by each transformation.

The evidence unit is a design decision, not a neutral preprocessing step. Fixed-size chunks are convenient for vector indexes but can split definitions from conditions, claims from qualifiers, and table captions from tables. Whole documents preserve context but exceed retrieval and generation budgets. Sections, paragraphs, propositions, claims, and figure-table units each create different recall and attribution properties. Recent studies continue to find that chunking materially changes RAG performance, and hierarchical or context-preserving strategies are increasingly used to avoid isolated fragments [47–49]. For scientific automation, the correct unit may be multi-layered: a retrievable claim linked to its paragraph, section, document, version, and provenance graph.

The final distinction is between retrieval quality and answer quality. A retriever can return the gold evidence and the generator can still fail. A generator can answer correctly from parametric memory despite failed retrieval. End-to-end accuracy alone hides both cases. Conversely, a retrieval benchmark can reward passages that contain the answer string while ignoring whether the downstream task requires comparison, contradiction, or causal interpretation. A mature RAG programme measures the stages separately and then studies their interaction.

This decomposition changes how RAG should be assessed. The relevant question is whether a specified pipeline, corpus, task distribution, and budget produces better decisions than specified alternatives; whether its provenance is sufficient for audit; and whether it responds appropriately when evidence is missing. The remainder of the first part develops the architecture and evaluation concepts required to answer those questions.

Chapter 2. Sparse, Dense, Hybrid, and Late-Interaction Retrieval

The retrieval stack begins before any index is built. Source acquisition determines the ceiling of recall. A system cannot retrieve an article it never ingested, a table it failed to parse, or a correction whose identifier was not connected to the original. In research environments, acquisition must cover both conventional publications and the artefacts that make results interpretable: supplementary files, code repositories, datasets, protocols, registered reports, review histories, errata, retractions, and versions. The ingestion pipeline should preserve stable source identifiers and content hashes so that later answers can be reproduced against a known snapshot.

Parsing has semantic consequences and should not be treated as routine file conversion. PDF text extraction may scramble reading order, drop equations, detach captions, flatten tables, and merge headers into body text. Optical character recognition can introduce character substitutions that are consequential for gene names, equations, identifiers, or chemical formulas. A sound pipeline retains the original file, an extracted representation, structural coordinates, and uncertainty about extraction. When possible, it stores the relation between a normalized evidence unit and its exact page or source span. This supports both citation display and verification against the material available to the user.

Segmentation creates the retrieval units. The simplest pattern uses fixed token windows with overlap. It is easy to implement and often competitive, but its apparent neutrality is misleading. Small chunks improve localization and reduce irrelevant context, yet they may omit necessary qualifiers. Large chunks preserve coherence but dilute embeddings and consume context. Overlap improves recall at the price of duplication, which can dominate a top-k result and create a false impression of multiple independent sources. Section-aware segmentation, semantic boundary detection, sentence windows, parent-child retrieval, and late chunking attempt to preserve context without indexing entire documents [47–49]. The appropriate strategy depends on the evidence granularity required by the task.

Sparse retrieval remains a core component of modern retrieval systems. BM25 estimates relevance from query-term occurrences while controlling for document length and term-frequency saturation [8]. It is fast, interpretable, and robust under domain shift. BEIR found BM25 to be a competitive zero-shot baseline across heterogeneous retrieval tasks, while late-interaction and reranking systems often achieved the best average quality at higher computational cost [9]. Sparse retrieval is particularly valuable for exact terminology, identifiers, citations, legal phrases, abbreviations, formulas represented as text, and names that a dense model may map poorly. It should be retained in most production systems unless a task-specific evaluation demonstrates that it adds no value.

Dense retrieval maps queries and documents into learned vector spaces. Dual encoders make document embeddings precomputable, enabling approximate nearest-neighbour search at scale [2]. Dense models can retrieve paraphrases and

conceptually related passages without shared vocabulary, but they inherit the distribution and objectives of their training data. A general embedding model may conflate topical similarity with task relevance, miss rare technical distinctions, or rank fluent background above the exact methodological evidence required by a scientific question. MTEB and its multilingual expansion show that no embedding approach dominates across all tasks, languages, and domains [10]. Model selection must therefore follow the target retrieval distribution, not a leaderboard headline.

Learned sparse retrieval occupies an intermediate space. Models such as SPLADE expand or reweight lexical representations while retaining inverted-index efficiency. They can capture semantic associations without abandoning term-level explainability and sparse infrastructure. In practice, many strong systems use hybrid retrieval rather than committing to one representation. Sparse and dense rankings can be combined by weighted scores, reciprocal-rank fusion, learned fusion, or conditional routing. Hybridization is not a guarantee; badly calibrated score fusion can suppress the strongest channel. The system should report channel-specific recall and the marginal contribution of each retriever.

Late interaction addresses the information loss of a single vector. ColBERT encodes query and document tokens separately and computes a MaxSim-style score in which each query representation interacts with its best matching document representation [6]. ColBERTv2 improves storage and supervision while preserving this fine-grained interaction [7]. The practical interpretation is important: a document can be relevant because several distinct query facets match different local regions, even if no single pooled embedding represents the relationship well. This makes late interaction a strong baseline for any EPR claim about multifaceted relevance. If a proposed portfolio method only succeeds because a single-vector retriever blurred the query, ColBERTv2 may remove the apparent advantage.

Cross-encoders provide stronger but more expensive relevance estimation. They jointly encode a query and candidate text, allowing full token-level attention. Because every pair requires inference, cross-encoders normally rerank a candidate pool rather than search the entire corpus. The common design is retrieve broadly with sparse, dense, or late-interaction methods, then rerank tens or hundreds of candidates. The candidate stage optimizes recall; the reranker optimizes precision. This separation is one of the most reliable RAG design patterns. It also exposes a hard limit: no reranker can recover evidence absent from the candidate pool.

Candidate generation can be expanded through query transformation. The system may generate paraphrases, technical synonyms, controlled vocabulary terms, citations, or hypothetical documents. It may decompose a complex question into subqueries and union the results. These techniques improve recall when the user's phrasing differs from the corpus, but they also widen the attack and error surface. A generated expansion may encode a false assumption and retrieve documents that reinforce it. Robust implementations keep transformations independent long enough to compare their results, attach provenance to each candidate, and prevent an unverified expansion from silently replacing the original information need.

Metadata filtering is another retrieval channel. Date, jurisdiction, study design, population, publication status, access rights, author, venue, dataset, and version may

matter more than semantic similarity. Scientific questions frequently require temporal cut-offs or distinctions between preprints and peer-reviewed work. Filters should be applied with awareness of recall. A missing date should not always exclude a document; an incorrect date should not silently admit it. Metadata quality therefore needs its own audit and uncertainty model.

Index architecture determines operational properties. Inverted indexes support fast sparse search and incremental updates. Vector indexes use structures such as HNSW or inverted files to approximate nearest neighbours. Late-interaction indexes store multiple compressed vectors per document. Graph indexes encode explicit relations. A mature system often maintains several indexes over the same versioned evidence units. Their synchronization is part of correctness. If a document is retracted in the metadata store but remains retrievable from a stale vector index, the logical corpus and the operational corpus differ.

Reranking should not be confused with evidence selection. A reranker estimates the order of individual candidates. A selector may need to choose a set under a token budget, remove near duplicates, preserve source independence, or satisfy role constraints. MMR introduces a simple iterative trade-off between query relevance and redundancy [23]. Intent-aware diversification methods such as IA-Select and xQuAD allocate results across possible subtopics [24–25]. Determinantal point processes and submodular objectives provide more general set-selection frameworks [26]. These methods are predecessors of EPR, but they normally optimize diversity or intent coverage rather than explicit scientific evidence roles.

The final output of the retrieval stack should be richer than a list of strings and scores. Each candidate should carry its source identifier, version, retrieval unit, parent context, channel scores, transformation provenance, access policy, timestamp, and any structural links. For scientific use, it should also carry publication type, study design where known, citation relationships, and independence warnings. These fields are not decoration. They become inputs to context construction, portfolio selection, citation verification, and downstream reasoning.

A practical retrieval stack is best designed as a recall-preserving funnel. The corpus and parsing layers preserve potentially useful evidence. Multiple retrieval channels create a candidate pool. Rerankers estimate local relevance. Selectors construct a budgeted context. Every narrowing step should be measurable, reversible in analysis, and attributable. Premature pruning is a central cause of failure in multi-hop and research-oriented systems; NEST's separation of recall amplification from precision selection is a recent expression of this principle [28]. Precision can be improved at later stages, whereas discarded evidence cannot participate in subsequent reasoning.

Chapter 3. Context Construction, Grounding, and Generation

Retrieval returns an evidence proposal rather than a completed answer. The generator must still determine which passages matter, reconcile or expose conflicts, preserve qualifications, and attach claims to sources. Many RAG failures occur after successful retrieval because context construction and generation are treated as prompt formatting rather than as engineered stages with explicit contracts.

Context construction begins with budgeting. The relevant budget is not only the model's advertised context window. It includes the number of tokens that can be processed at acceptable latency and cost, the amount the model can use reliably, and the space required for instructions, conversation history, tool traces, and output. "Lost in the middle" experiments showed that models can use long contexts unevenly, with performance depending on the position of relevant information [15]. Later systems have improved, but order, duplication, and distractor density still affect behaviour. Simply increasing the number of retrieved passages can reduce answer quality.

A context builder decides what to include, how to order it, and at what granularity. A common pattern places the highest-ranked evidence first, but this can overemphasize a single perspective. Another groups passages by source or facet. A scientific system may order evidence by claim, chronology, study design, or contradiction structure. The ordering should reflect the downstream task rather than a generic relevance score. A novelty analysis may begin with the closest holistic precedent, then component precedents, then boundary cases. A peer-review context may pair supporting and contradicting evidence so that the generator cannot read one without the other.

Deduplication is essential. Chunk overlap, mirrored repositories, preprint and final versions, review articles that quote primary studies, and syndicated web content can produce many apparently independent passages that derive from the same source. A pipeline that does not model source dependence may interpret repetition as evidential strength. In scientific use, independence matters. A context constructor should recognize exact duplicates, near duplicates, version families, citation-derived repetition, and common upstream sources. RARE's emphasis on high-similarity corpora illustrates how conventional retrieval metrics can conceal robustness problems when redundant evidence dominates [31].

Context compression reduces cost and distraction, but it changes the evidence. Extractive compression selects sentences or spans. Abstractive compression rewrites them. Structured compression maps text into claims, entities, relations, or tables. Each choice creates a different verification burden. Extractive spans preserve wording but may omit context. Abstractive summaries can merge claims or strengthen modality. Structured representations are useful for computation but depend on a semantic compiler. For high-stakes scientific use, the compressed unit should retain a link to the original passage and a record of the transformation.

Generation architectures differ in how they consume evidence. In prompt-based RAG, retrieved passages are concatenated with instructions and fed to a general model. FiD-style models encode passages separately and allow the decoder to aggregate them [4]. Retrieval-enhanced language models such as RETRO integrate neighbours during model computation [5]. Agentic systems interleave search, reading, drafting, and verification. The architecture affects error attribution. In prompt-based systems, retrieval can be inspected independently. In tightly integrated systems, retrieval and generation may be jointly optimized but harder to audit.

The generator’s instruction should define an evidence policy. It should state whether external evidence overrides parametric memory, how conflicts are reported, when the system must abstain, and what counts as a supported claim. These policies should not rely solely on natural-language prompt compliance. Structured output contracts, claim-level citation requirements, and post-generation verification provide stronger controls. The generator should be permitted to say that the retrieved portfolio is insufficient, rather than being forced to synthesize a definitive answer from whatever context was supplied.

Citation generation is distinct from citation correctness. A model can attach a relevant-looking source to a claim that the source does not entail. It can cite the right paper for the wrong sentence, cite a review when a primary source is required, or omit the passage that supports a quantitative value. Recent work on agentic scientific synthesis found that measured unsupported-citation rates vary substantially with verifier strictness and argued that citation evaluation must name and calibrate the verifier rather than treating automated attribution as ground truth [45]. In EPR, citation support is a defined requirement of the portfolio contract.

Conflict handling is another distinct stage. Retrieved documents may contradict one another because one is false, because methods differ, because populations differ, because terminology is ambiguous, or because the literature evolved. The generator should not average these differences into a smooth paragraph. It should identify the scope of each claim, expose incompatible evidence, and preserve uncertainty. Work on RAG under context-memory and inter-document conflict shows that models can ignore external evidence, blend incompatible sources, or follow misleading context [42–44]. A scientific assistant therefore needs explicit conflict representations and, ideally, separate retrieval for adjudicating evidence.

Grounded generation also requires negative evidence discipline. The absence of a retrieved supporting passage does not imply that a claim is false. The absence of a contradicting passage does not imply consensus. A failed search may reflect a narrow query, missing corpus, language mismatch, paywall, indexing lag, or terminology drift. The output should distinguish a supported positive claim, supported negation, conflicting evidence, and insufficient search coverage. These states are more useful than a single confidence score.

Long-form synthesis introduces additional problems. Local sentence faithfulness can coexist with global distortion. A report may cite every paragraph correctly while allocating disproportionate space to one line of evidence, omitting negative studies, or presenting an unresolved debate as settled. OpenScholar demonstrates the potential of large-scale scientific retrieval and citation-backed synthesis over tens of millions of papers [22].

PaperQA2 reports strong results on literature search, summarization, and contradiction detection [21]. At the same time, benchmarks and citation-audit work show that retrieval coverage, attribution, and judge calibration remain central bottlenecks [33,45]. The engineering objective therefore includes both claim-level citation support and the sufficiency and representativeness of the evidence set.

Verification can be layered. A retrieval verifier checks whether selected passages are relevant. An entailment verifier checks whether each citation supports its claim. A consistency verifier checks whether claims conflict. A provenance verifier checks source identity and version. A task verifier checks whether the response answered the requested question. A portfolio verifier checks coverage of required evidence roles. These layers may use models, rules, human review, or combinations. Their error rates must be measured separately; a weak verifier can create the illusion of a trustworthy pipeline.

The output interface should expose enough structure for correction. A user should be able to inspect the passages behind a claim, replace a source, mark two sources as dependent, reject an interpretation, and request search for a missing facet. Corrections should alter persistent evidence metadata or task traces, not merely the wording of one answer. This is especially important in a research IDE, where the same corpus supports novelty assessment, peer review, compliance, and experiment planning.

The transition from retrieval to generation is therefore an engineered system boundary rather than only a prompt-composition step. It is where an independently ranked list becomes an evidence object for a downstream decision. Conventional RAG often delegates this transformation to the generator. EPR makes part of it explicit by constructing a portfolio before generation. Whether the additional layer justifies its cost remains an empirical question developed in the second half of the book. Retrieval success is necessary but insufficient: evidence must survive selection, remain distinguishable, and be consumed under a policy appropriate to the scientific task.

Chapter 4. Advanced Retrieval-Augmented Architectures

The state of the art comprises several architectural patterns for deciding when, where, and how retrieval participates in computation. Since the original RAG formulation, retrieval has expanded beyond a single top-k operation performed once before generation. Modern systems route queries, invoke several retrievers, construct hierarchical representations, traverse graphs, retrieve iteratively, and evaluate their own evidence. The engineering objective is to select the least complex architecture that addresses the observed task requirements and failure modes.

Hybrid RAG is the default practical pattern. Sparse retrieval preserves exact lexical evidence, while dense retrieval provides semantic recall. Late-interaction models add fine-grained matching, and cross-encoders improve final ranking. Reciprocal-rank fusion is often preferred to raw-score averaging because retrieval channels produce scores on incompatible scales. A production pipeline may first union candidates from BM25, a domain-tuned dense model, and a citation or metadata search, then rerank the union. The gain comes less from an abstract notion of hybridity than from error diversity: a rare acronym missed by a dense model may be obvious to BM25, while a paraphrased mechanism missed by lexical search may be recovered by a dense encoder.

Adaptive RAG changes whether and how much retrieval is used. Not every query requires external evidence, and repeated retrieval can waste latency or introduce noise. Self-RAG trains a model to retrieve, generate, and critique through special reflection decisions, allowing retrieval to be invoked selectively and outputs to be assessed for support [12]. Related approaches use uncertainty, query complexity, hallucination detection, or learned routers to choose among no retrieval, single-step retrieval, multi-step retrieval, and tool use. The architectural lesson is that retrieval policy is itself a modelled decision. The policy must be evaluated for both false negatives, where retrieval was needed but skipped, and false positives, where unnecessary retrieval harmed a correct parametric answer.

Corrective and reflective RAG add a feedback loop after initial retrieval. The system evaluates whether evidence is relevant or sufficient, rewrites the query, broadens sources, or falls back to a web search. The intuitive design is attractive, but self-evaluation can be circular: the same model that misunderstood the question may judge its own evidence as adequate. Strong implementations separate roles, use independent signals, or evaluate the loop against an oracle decomposition. Each additional round should have a measurable stopping rule, because unconstrained iterative retrieval can accumulate distractors faster than useful evidence.

Multi-hop RAG addresses questions whose answer depends on evidence distributed across sources. HotpotQA made this requirement explicit in question answering [40], while MultiHop-RAG and later benchmarks show that standard pipelines often fail because the first query does not mention the entity needed for the second retrieval step [17]. Query decomposition generates intermediate questions. Iterative retrieval uses an

answer or entity from one hop to formulate the next. Learned multi-hop retrievers attempt to model the path directly. In 2026, systems such as GRITHopper and structured program-guided retrieval continue to reduce dependence on fragile autoregressive decompositions. The core engineering problem is evidence survival: an apparently marginal passage may become decisive only after another passage establishes the bridge.

Hierarchical RAG changes the granularity of the index. RAPTOR recursively clusters and summarizes text, creating a tree whose upper levels support thematic or global queries and whose leaves preserve local evidence [13]. Parent-child retrieval indexes small units for recall but returns larger parent contexts for generation. Document-summary indexes first identify relevant documents, then search within them. Multi-granular systems route among sentence, paragraph, section, document, and corpus-level representations. These patterns address the mismatch between a short retrieval unit and a broad information need, but they also introduce derived summaries whose errors can propagate. The index is no longer only a copy of the corpus; it contains model-generated abstractions that require versioning and provenance.

Graph RAG encodes relationships among entities, claims, passages, or documents. Microsoft’s GraphRAG extracts an entity-relation graph, detects communities, and generates community summaries to answer global questions about a corpus [14]. The method is valuable for questions concerning dominant themes, relationships, or narratives across a large collection, where nearest-neighbour retrieval of local chunks is structurally mismatched. Knowledge-graph RAG retrieves explicit paths or subgraphs for multi-hop reasoning. Graph construction can nevertheless be expensive and error-prone. Entity-resolution failures create false edges; extraction omissions remove paths; and community summaries can become stale when documents change. Graph-based retrieval is most appropriate when relationships and global aggregation are explicit task requirements.

Subgraph retrieval is a particularly relevant neighbouring field for EPR. COSMOS formalizes the selection of coherent evidence subgraphs as constrained submodular maximization, explicitly optimizing collective utility and connectivity instead of treating facts independently [29]. This is close in spirit to portfolio retrieval, though its primary setting is knowledge-base question answering rather than scientific evidence roles. The distinction matters: connectivity can indicate a valid reasoning path, but a scientific portfolio may deliberately include disconnected counterevidence or an analogy from another domain. EPR can borrow the set-level optimization idea without assuming that all useful evidence belongs to one connected graph.

Global and local retrieval can be combined. A system may first use corpus-level summaries to identify themes, then retrieve local passages to support claims. This pattern is useful for literature mapping and research landscape analysis. It is less appropriate for exact factual questions because the additional summarization layer increases cost and creates another source of abstraction error. Evaluations of GraphRAG variants increasingly show that gains depend strongly on query type, graph quality, and baseline strength. “Graph RAG” is therefore a family of conditional design choices rather than a generally superior successor to vector search.

Agentic RAG exposes retrieval operations to a model as tools. The agent decides which corpus to search, which query to issue, which result to open, whether to inspect references, and when to stop. This resembles an iterative research workflow and is useful when the search strategy depends on intermediate findings. PaperQA2 and OpenScholar demonstrate the value of repeated search, reranking, evidence reading, and citation-backed synthesis in scientific literature tasks [21–22]. Agentic retrieval also introduces loops, unbounded cost, source drift, query contamination by earlier model inferences, and weak reproducibility. Operational systems should record each action and support replay against a fixed corpus snapshot.

Multimodal RAG retrieves not only text but tables, figures, diagrams, page images, audio, or video. Scientific evidence is often multimodal: a conclusion may depend on a plot, a microscopy image, a table footnote, or a mathematical derivation. Text extraction alone can remove decisive information. Multimodal indexes can operate over page images or modality-specific representations, while graph systems connect figures to captions and surrounding text. UniversalRAG and 2026 multimodal graph approaches reflect the move toward routing queries across heterogeneous corpora. The difficulty is evaluation: a system may retrieve the correct page but fail to interpret the figure, or interpret the figure while citing a text fragment that does not support the conclusion.

Temporal RAG treats time as part of relevance. The same query may require the most recent guideline, the state of knowledge before a patent filing, or the literature available when a study was designed. Indexes need valid-time and ingestion-time metadata. Retrieval should prevent future leakage in retrospective evaluations. Scientific novelty is inherently temporal: an idea may have been novel in 2022 and routine in 2026. Any EPR system for prior art or novelty must define the cut-off date as part of the task contract.

Long-context models complicate the architecture choice. If an entire corpus fits in context, retrieval may appear unnecessary. Benchmarks comparing long-context models and RAG show no universal winner; the result depends on corpus length, task, evidence distribution, model, and cost [15,46,50]. Long context avoids retrieval misses but increases latency, cost, distractor exposure, and positional sensitivity. RAG reduces the material presented to the model but risks excluding evidence. Hybrid designs use retrieval to organize long contexts, preserve source boundaries, or select a subset for expensive reasoning. The relevant comparison is not “RAG versus context window” in the abstract but quality per unit of cost, with realistic document distributions.

Memory-oriented RAG extends retrieval across sessions. A system may index user interactions, corrections, plans, or prior reasoning traces. This is useful for research programmes that evolve over months, but conversational memory differs from scientific evidence. A user preference, an unverified hypothesis, and a published result should not inhabit the same undifferentiated index. Memory systems need typed stores, retention policies, access controls, and rules for superseding earlier statements.

The pattern that unifies these architectures is control over information flow. Hybrid retrieval controls representation. Adaptive RAG controls invocation. Multi-hop RAG controls sequencing. Hierarchical and graph RAG control structure. Agentic RAG controls search strategy. Multimodal RAG controls evidence type. None of these removes the need for strong baselines. Each architectural addition should correspond to an observed

failure class and should survive an ablation in which corpus, generator, and budget are held constant.

EPR must be evaluated against a mature modern RAG stack. A credible comparison includes broad candidate retrieval, query decomposition, high-quality reranking, source deduplication, and result diversification. The EPR hypothesis is narrower: after these capabilities are present, explicit portfolio roles and evidence-sufficiency objectives may still improve downstream scientific decisions. The following chapters specify the failure classes and task conditions under which this hypothesis can be tested.

Chapter 5. Failure Modes Across the RAG Pipeline

RAG failures are often reported as hallucinations, but that label is too coarse for engineering. A false answer can result from a missing source, a broken parser, an inappropriate chunk, a retrieval miss, a bad reranking decision, context overload, an ignored passage, a faulty synthesis, an unsupported citation, or an adversarial document. Treating all of these as generator hallucination prevents targeted improvement. This chapter organizes failures along the information path and distinguishes defects that EPR may address from those it cannot.

Corpus incompleteness is the first failure. The relevant evidence may not be indexed because it is paywalled, unpublished, in another language, stored in supplementary material, or absent from the chosen repository. In novelty assessment, incomplete coverage is especially dangerous because “not retrieved” is easily misread as “not known.” A system must report corpus scope and search coverage. It should distinguish the failure to find a close precedent from evidence that no precedent exists. No retrieval algorithm can compensate for an unsearched corpus.

Corpus bias is subtler. A collection may overrepresent highly cited work, English-language publications, positive results, accessible journals, specific regions, or one methodological school. Retrieval then amplifies the collection’s bias while appearing evidence-based. In peer review, this can create a consensus illusion. In scientific discovery, it can suppress anomalies and negative findings. Portfolio retrieval can mitigate redundancy within the corpus but cannot manufacture missing perspectives. Corpus audits and source diversification remain necessary.

Source-state failures arise when versions, corrections, and retractions are mishandled. A preprint and final article may be retrieved as independent support. A retracted result may outrank its notice. A corrected table may coexist with the original. The source layer must model version families and status changes. A generator should cite the exact version used. Scientific automation without version semantics is not reproducible.

Parsing failures alter content before retrieval. Multi-column PDFs can interleave text. Equations can disappear. Tables can be flattened into nonsensical sequences. Figure references can detach from images. Headers and references can become chunks that dominate lexical search. These defects are visible only if the pipeline preserves structural diagnostics and sample-based quality checks. Improvements in retriever architecture do not fix corrupted evidence.

Segmentation failures occur when the evidence boundary does not match the indexed unit. A claim may be in one chunk and its negation or condition in another. A method description may be separated from the population on which it was evaluated. Overlap can create duplicates; no overlap can create broken context. Chunking should be benchmarked against source-span retrieval, not chosen by convention. Hierarchical and

parent-child patterns reduce some failures, but they also increase implementation complexity and context size.

Representation failure occurs when the retrieval model encodes the wrong notion of similarity. BM25 can miss paraphrases. Dense models can miss exact rare terms or overvalue topical similarity. Single-vector models can blur multiple facets. Late interaction can overmatch locally related words without capturing a global relation. Domain shift degrades learned models unpredictably; BEIR and MTEB were created partly because performance on one benchmark does not imply robust transfer [9–10]. The practical response is multi-channel retrieval and domain-specific evaluation.

Query-understanding failure occurs when the system retrieves for the literal wording rather than the actual information need. Ambiguous entities, hidden time constraints, comparative questions, and multi-part requests are common. Query expansion can help but may introduce assumptions. For example, expanding “model robustness” into “adversarial robustness” can exclude distribution-shift evidence. The original query and transformed queries must remain separately inspectable.

Recall failure is the absence of necessary evidence from the candidate pool. It is the most consequential retrieval error because later stages cannot recover discarded items. Multi-hop tasks are particularly vulnerable: the relevance of a passage may be visible only after another passage reveals a bridge. NEST identifies premature pruning as a major source of recall loss and evaluates candidates under nested contexts to preserve evidence that looks weak in isolation [28]. EPR inherits the same constraint; a portfolio selector cannot select what the retrieval funnel removed.

Precision failure is the presence of irrelevant or weakly relevant evidence. Some noise is tolerable in the candidate pool but harmful in the final context. Dense retrievers may retrieve conceptually adjacent background rather than the specific claim required. LLM query expansion can generate fluent but misleading pseudo-documents. A strong reranker reduces this noise, but its scores may be poorly calibrated across domains or source types.

Redundancy failure occurs when the top results repeat the same evidence. Overlapping chunks, duplicate versions, survey quotations, and papers from one research lineage can consume the context budget. Standard ranking metrics may reward these results if each is individually relevant. RARE shows how retrieval performance can collapse in realistic high-similarity corpora even when conventional evaluation appears strong [31]. Redundancy is the clearest failure class that motivates set-level selection.

Coverage failure is different from redundancy. The results may be diverse but omit a required facet. A novelty query may retrieve work on peer review and provenance but nothing on human oversight. A causal question may retrieve mechanism and outcome but no intervention evidence. Generic diversity does not guarantee task-relevant coverage. EPR proposes explicit facet and role coverage, but the facets themselves must be correctly inferred and should not force irrelevant documents into the portfolio.

Independence failure occurs when apparently separate sources are not independent. Several papers may reuse the same dataset, codebase, or upstream claim. A review article and the primary study it summarizes should not count as two independent confirmations. Citation graphs, shared authors, dataset metadata, and textual provenance

can provide dependency signals. The system should expose them rather than silently converting document count into evidence strength.

Temporal failure occurs when retrieval ignores time. A current answer may cite an obsolete standard. A prior-art analysis may leak documents published after the relevant filing date. A systematic review may mix literature outside its protocol window. Temporal constraints must be applied at retrieval and verified after selection because metadata can be missing or wrong.

Source-quality failure occurs when relevance overwhelms authority or methodological quality. A blog post may explain a concept better than a paper and therefore rank higher, even when the task requires primary evidence. A highly cited paper may still have weak methods. Retrieval should not collapse relevance, authority, and evidential strength into one score. These are separate dimensions. For scientific use, source-type and quality metadata should accompany rather than replace textual relevance.

Context-construction failure occurs when good evidence is packaged badly. Relevant passages can be truncated, ordered so that the model anchors on one source, or buried among distractors. Duplicate evidence can dominate attention. The same retrieved set can produce different answers under different permutations; 2026 work on retrieval-order sensitivity continues to show that context ordering is not a neutral operation [51]. Context assembly needs deterministic policies and permutation tests.

Context-budget failure occurs when the system retrieves more evidence than the generator can use. Advertised context length is not equivalent to effective evidence capacity. Long contexts increase cost and can reduce utilization. Compression may omit qualifiers. A portfolio approach addresses budget allocation directly, but it cannot eliminate the fundamental trade-off between coverage and cognitive load.

Evidence-use failure occurs when the generator ignores relevant context. Models may answer from parametric memory, especially when external evidence conflicts with familiar facts. They may copy an incorrect passage because it is salient or follow a majority of redundant sources. The presence of a gold document in context is not proof that it influenced the answer correctly. Stage-specific evaluation must separate retrieval, utilization, and generation.

Synthesis failure occurs when the generator combines individually supported statements into an unsupported conclusion. This is common in multi-paper scientific summaries. One paper establishes A, another establishes B, and the model asserts A causes B. Claim-level citation checks may pass for the premises while the synthesis step remains invalid. EPR can provide more complete evidence, but only an explicit reasoning or verification layer can assess the inference.

Conflict-smoothing failure occurs when the model resolves disagreement through rhetoric rather than evidence. It may write that “results are mixed” when studies concern different populations, or present a median view that no source supports. Work on conflicting evidence shows that RAG systems need explicit mechanisms for ambiguity, misinformation, noise, and context-memory conflict [42–44]. A research assistant should preserve the structure of disagreement.

Citation failure includes missing, irrelevant, overbroad, or misplaced citations. A source may be topically related but not entail the sentence. A single citation may be

attached to a paragraph containing several unsupported claims. Automated verifiers differ in strictness, so reported citation accuracy is partly an instrument property [45]. Citation audits should operate at claim or atomic-statement level and identify the verifier protocol.

Calibration failure occurs when system confidence does not reflect evidence quality. A fluent answer with weak retrieval may receive high confidence. A correct answer based on an exact clause may be hedged. Confidence should incorporate retrieval coverage, conflict, source quality, and verifier results rather than model self-report alone. EPR similarly requires calibrated sufficiency: a portfolio must be able to declare that a required role is missing.

Evaluation leakage creates false progress. Synthetic questions generated from indexed passages can make retrieval trivial. Benchmarks may overlap model training data. Gold answers may be recoverable from parametric memory, masking retrieval failure. A benchmark may reward one wording of evidence while penalizing a valid alternative. Fresh, temporal, blind, and adversarial evaluations are required. BRIGHT demonstrates how dramatically retrieval performance can fall when relevance requires reasoning rather than surface matching [16].

Security failures extend the taxonomy beyond accidental error. Retrieval poisoning injects documents designed to be retrieved and influence answers. Indirect prompt injection embeds instructions in retrieved content. Attacks can target particular queries, exploit trust in citations, or combine poisoning with prompt injection [52–54]. The corpus must be treated as untrusted input. Retrieved text should not be allowed to redefine system instructions, invoke tools, or alter access controls.

Privacy and access-control failures occur when retrieval surfaces information a user is not authorized to see, or when embeddings and caches leak sensitive content. Filtering after retrieval may expose titles, snippets, or timing signals. Permissions should be enforced before candidate material enters the model context. Scientific systems must also respect embargoes, participant privacy, contractual restrictions, and data-use constraints.

Operational failures include stale indexes, inconsistent replicas, nondeterministic reranking, missing observability, and cost explosions under iterative search. A system can be accurate in a notebook and unreliable in deployment because timeouts alter the retrieval path. Every stage should emit traceable metrics: candidate counts, channel latency, index version, reranker version, context tokens, generator configuration, and verifier results.

EPR addresses only a subset of this taxonomy. It is primarily aimed at redundancy, facet coverage, role coverage, source independence, context budgeting, and explicit sufficiency. It may help expose conflicts and gaps. It does not solve corpus incompleteness, parsing corruption, bad embeddings, false source metadata, generator hallucination, or poisoning by itself. This boundary is essential. A new selection objective is valuable only when built on a competent retrieval and governance stack.

Chapter 6. Evaluation Methodology and Benchmark Design

Evaluation is where RAG research most often loses engineering meaning. A pipeline has several stages, each with different metrics, yet a single answer score is frequently used to justify architectural claims. The opposite mistake is to optimize retrieval metrics that do not predict downstream usefulness. A rigorous programme needs a measurement model that links corpus coverage, retrieval, context construction, evidence utilization, generation, citation, cost, and human decision quality.

Traditional information-retrieval metrics remain foundational. Precision@k measures the proportion of returned items judged relevant. Recall@k measures the proportion of relevant items recovered within the first k. Mean reciprocal rank rewards placing the first relevant item early. Average precision summarizes ranking quality across relevant items. Discounted cumulative gain assigns graded relevance and discounts lower ranks; normalized DCG compares the ranking with an ideal ordering. These metrics are interpretable only relative to a relevance definition and a judgement pool. In research tasks, there may be many valid evidence sets and no complete enumeration of relevant documents.

Recall is the critical candidate-generation metric. If the final answer requires three evidence units and only two reach the pool, reranking and generation cannot repair the omission. Multi-hop evaluation should therefore measure complete-evidence recall, not merely whether any relevant passage appears. RARE introduces PerfRecall to expose failures under redundant, multi-hop corpora [31]. EPR extends the concern from document recall to role and facet recall: did the pool contain at least one credible candidate for each evidence function required by the task?

Ranking metrics such as nDCG are necessary but insufficient for portfolio selection. A diversified or role-aware selector may intentionally rank a moderately relevant but complementary document above a near duplicate. Conventional nDCG can penalize that choice if relevance labels are document-centric. The solution is not to discard nDCG. It is to report it alongside set-level metrics and downstream performance. A method that gains coverage by filling the context with peripheral material should not escape the precision penalty.

Retrieval unit evaluation must match the evidence unit. A system may retrieve the right paper but the wrong passage. Document-level recall can hide passage-level failure. Conversely, a correct paragraph may be retrieved from a secondary source when the task requires the primary study. Evaluation should distinguish source discovery, evidence localization, and source suitability. For multimodal evidence, page, figure, table, and cell-level measures may be required.

Generation metrics depend on the task. Exact match and token F1 work for short answers but are weak for synthesis. ROUGE and semantic similarity can reward fluent paraphrases without factual support. Long-form answers require claim-level correctness, completeness, relevance, coherence, and calibration. CLAPNQ and Long2RAG reflect

efforts to create more realistic long-form evaluation settings [46,55]. For scientific synthesis, citation correctness and evidence representativeness are as important as textual quality.

RAGAS decomposes automated evaluation into measures such as faithfulness, answer relevance, context relevance, and context recall without always requiring reference answers [18]. ARES trains lightweight judges from a small number of human labels and synthetic examples to evaluate context relevance, answer faithfulness, and answer relevance [19]. MIRAGE expands metric-intensive benchmarking [20]. These frameworks are useful for iteration, but LLM-based evaluators are not neutral instruments. Their prompts, models, thresholds, and domain transfer must be validated against human gold. The citation-faithfulness results discussed earlier show how much measured error can vary with verifier strictness [45].

Benchmark families test different parts of the stack. BEIR evaluates zero-shot retrieval across heterogeneous domains and shows the importance of robust baselines [9]. MTEB and MMTEB evaluate embeddings across tasks and languages rather than assuming one universal representation [10]. KILT links knowledge-intensive tasks to provenance in a shared Wikipedia snapshot [41]. HotpotQA tests multi-hop question answering [40]. SciFact and FEVER evaluate claim verification with evidence [38–39]. BRIGHT tests reasoning-intensive retrieval and reveals a large gap between conventional embedding performance and queries whose relevance must be inferred [16]. MM-BRIGHT extends this challenge to multimodal retrieval [56]. No single benchmark represents scientific EPR, but together they prevent an implementation from overfitting to one easy regime.

Scientific retrieval requires specialized benchmarks. LitQA2 evaluates realistic literature questions used in the PaperQA2 programme [21]. OpenScholar evaluates scientific synthesis over a very large open-access corpus [22]. RAG4SR-CS-200 provides protocol-oriented systematic-review artefacts spanning retrieval, screening, citation-grounded generation, and structured tables [35]. IntraBench evaluates content-grounded information retrieval through literature review [36]. PaperScope introduces multi-document scientific tasks across retrieval, reasoning, summarization, and problem solving [37]. These benchmarks increasingly recognize that scientific work is not reducible to retrieving an answer-bearing sentence.

Novelty assessment requires direct evaluation against expert judgement. RINoBench contains 1,381 research ideas with human novelty judgements and finds that LLM rationales can resemble expert reasoning without producing reliably aligned novelty scores [32]. NovBench evaluates generated novelty assessments using paper-review pairs and criteria such as relevance, correctness, coverage, and clarity [33]. Patent retrieval can be evaluated with PatentEB, which includes patent-specific retrieval and embedding tasks at scale [34]. These resources are imperfect but essential. An EPR novelty system should not be validated only on synthetic concepts invented by its designers.

Peer-review evaluation is particularly difficult because a high-quality review is not unique. TreeReview structures review as a dynamic hierarchy of questions [57]. Benchmarks such as PRAIB and studies of automatic reviewers investigate reliability and behavioural failure. Work showing that automatic reviewers can miss faulty reasoning warns against evaluating only stylistic similarity to human reviews [58]. EPR should be

tested on whether it supplies the evidence needed to detect methodological flaws, unsupported claims, missing related work, and contradictions, not merely whether the final review sounds plausible.

Portfolio evaluation needs new but disciplined metrics. Facet Coverage asks what fraction of task-relevant aspects have credible evidence in the selected set. Role Coverage asks whether required epistemic roles are represented. Evidence Sufficiency asks whether an evaluator can make the target decision using only the portfolio. Redundancy measures semantic, source, and lineage duplication. Independence Coverage measures whether the portfolio contains genuinely distinct evidence origins. Context Efficiency divides downstream task quality by tokens, documents, latency, or cost. Portfolio Regret compares the selected set with an oracle set under the same budget. These metrics should be validated through human studies rather than accepted because they have intuitive names.

The primary EPR metric should be downstream decision quality at a fixed budget. For novelty, the decision may be a calibrated novelty classification plus a decomposition of known and uncovered contributions. For prior art, it may be claim-element coverage or the recall of novelty-destroying combinations. For peer review, it may be flaw detection, evidence-backed criticism, or expert preference. For scientific discovery, it may be hypothesis ranking, experimental discrimination, or the identification of missing mechanisms. The retriever is successful only if the portfolio improves the task that follows.

Budget control is non-negotiable. A portfolio system that retrieves thirty documents cannot be compared fairly with a top-k baseline that retrieves ten. The budget may be defined in documents, passages, tokens, model calls, wall-clock time, or monetary cost. Scientific systems often need a composite budget because one long paper can consume more context than several concise evidence units. Evaluation should include quality-budget curves rather than one arbitrary k.

Ablation is required for causal attribution. The same candidate pool should be passed to top-k relevance, MMR, xQuAD-style diversification, learned reranking, LLM selection, and EPR selection. The generator and prompt should remain fixed. Role labels can be removed to test whether they contribute. Source-dependency penalties, conflict search, and iterative gap retrieval should each be ablated. If EPR's gain disappears when a stronger reranker or query decomposition is used, the novelty claim must be narrowed.

Statistical significance should be accompanied by effect size and practical significance. Paired bootstrap or randomization tests can compare retrieval metrics. Human evaluations need inter-annotator agreement, blinded conditions, expertise reporting, and enough instances to estimate uncertainty. A two-point gain may be valuable in a high-stakes patent workflow or meaningless if it doubles cost. Confidence intervals should appear on all headline results.

Error analysis is as important as averages. Evaluation sets should be stratified by query type, evidence count, domain, temporal constraint, ambiguity, redundancy, and corpus coverage. EPR is expected to help most when evidence roles are distributed and the context budget is tight. It should not be presented as universally superior if gains are concentrated in that regime. Conversely, a neutral average can conceal a large benefit for the intended use case.

The experimental standard for this programme is therefore demanding. EPR must be compared with modern RAG, not with an embedding-only straw man. It must improve downstream scientific decisions at equal budget. It must preserve ordinary relevance and abstain when role coverage cannot be satisfied. It must expose which part of the gain comes from query decomposition, candidate recall, diversification, role classification, or portfolio optimization. The programme is successful only if the additional structure produces a result that a simpler pipeline cannot obtain reliably.

Chapter 7. Limits of Independent Top-k Ranking in Scientific Tasks

The central limitation of ordinary retrieval is not that relevance is unimportant. It is that relevance is usually assigned to documents one at a time, while difficult scientific decisions depend on the collective properties of a set. A document can be highly relevant yet add almost nothing after five near-duplicates have already been selected. Another document can be only moderately similar to the original wording yet supply the missing counterexample, methodological critique, enabling mechanism, or boundary condition that changes the conclusion. The engineering problem is therefore not always to rank individual documents more accurately. It is sometimes to assemble a constrained body of evidence whose members play complementary roles.

This distinction can be obscured by document-centric metrics and interfaces. A ranked list presents each result with a score, leaving the user to infer the value of the set. This is appropriate for simple lookup. If the question is “Which policy specifies a ninety-day retention period?”, the top result should contain the answer, and diversity has limited value. In research tasks, however, the question is often closer to “What should be considered before deciding whether this method is new, credible, safe, or worth testing?” The relevant output is an evidence configuration rather than a single passage.

Consider a proposal for an AI system that evaluates scientific manuscripts, extracts claims, verifies supporting sources, tracks provenance, checks reproducibility, and leaves the final judgement to a human reviewer. A strong top-k retriever may return several papers about automated peer review because that phrase dominates the query. The results can be individually relevant and collectively poor. They may omit work on claim verification, provenance systems, reproducibility tooling, contestability, safety cases, prior-art search, and human authority in automated decision systems. The missing documents are not random diversions. They provide the components, analogues, constraints, and counterevidence required to evaluate the proposal.

Two retrieval objectives

Individually relevant items versus a collectively useful evidence set

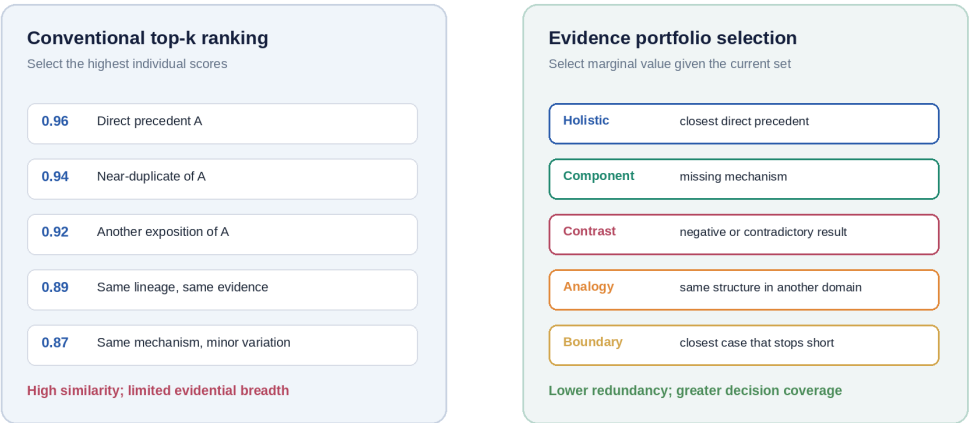


Figure 2. Top-k retrieval and portfolio retrieval optimize different objects. The latter values evidence for its marginal contribution to the current set.

This is particularly important in novelty assessment. A close holistic precedent can destroy a novelty claim, but the absence of such a precedent proves little. An idea may be unoriginal because its components and combination are distributed across several intellectual lineages. Conversely, a retrieved set can contain many papers sharing vocabulary while failing to explain the distinctive mechanism. A novelty workflow therefore needs at least three views of the literature: the closest complete precedents, precedents for the components and relations, and evidence about what remains unaccounted for after those precedents are considered. A single similarity ranking does not explicitly optimize any of these views.

Prior-art research exposes the same issue more sharply. Patent claims are combinations of elements, and potentially destructive prior art may be distributed across documents. Legal conclusions depend on jurisdiction, filing date, claim construction, and rules governing whether references can be combined. Retrieval cannot decide those questions. It can, however, avoid prematurely discarding references that cover less salient claim elements, earlier terminology, analogous mechanisms, or combinations that a later analyst must examine. The desired output is not simply “the ten nearest patents.” It is a compact search dossier with coverage across claim elements and plausible lines of combination.

Peer review has a different evidence structure. A reviewer evaluating a performance claim may need supporting studies, failed replications, benchmark limitations, alternative explanations, data-leakage risks, related methods, and the conditions under which the claimed effect disappears. A ranked list dominated by studies using the same terminology and benchmark can overrepresent one line of evidence. Retrieval redundancy becomes an epistemic problem when repeated material from one lineage is interpreted as independent confirmation. Portfolio construction should therefore account for source

families, citation dependence, dataset reuse, and methodological similarity in addition to textual diversity.

Scientific discovery adds another requirement: productive distance. A useful portfolio may include a close precedent, a mechanism from a neighbouring field, an anomaly unexplained by current theory, and a measurement technique capable of discriminating between hypotheses. The analogy may not rank highly for lexical or topical similarity. Its value arises from structural correspondence. This does not justify indiscriminate serendipity. It requires a disciplined representation of why a distant item might matter and a relevance floor below which diversity is not allowed to become randomness.

These examples reveal several meanings of relevance. Direct relevance measures whether a document addresses the expressed topic. Evidential relevance asks whether it changes the support for a claim. Procedural relevance asks whether it affects how a decision should be made. Contrastive relevance asks whether it distinguishes two hypotheses. Coverage relevance asks whether it represents an aspect missing from the selected set. Analogical relevance asks whether it instantiates a transferable structure. Temporal relevance asks whether it was available at the legally or scientifically relevant time. Independence relevance asks whether it adds a genuinely separate evidential origin. A top-k score usually collapses these meanings into one scalar trained for average ranking performance.

Modern RAG already contains partial responses to this problem. Query decomposition exposes hidden subquestions. Hybrid retrieval combines exact and semantic channels. Late interaction preserves local facets rather than compressing a document into one vector. Multi-hop systems retrieve iteratively. Graph retrieval follows dependencies. Maximal Marginal Relevance balances query similarity against redundancy [23]. xQuAD and intent-aware diversification represent subtopics explicitly [24–25]. Determinantal point processes model diversity as a property of a set [26]. These methods are important baselines, not predecessors to be dismissed.

The limitation is that diversity is only a proxy for evidential completeness. Two documents can be dissimilar and still serve no useful scientific role. A news article, an ethical essay, and a business analysis may diversify a biomedical search without helping to evaluate a mechanism. Conversely, two methodologically different replication studies can be semantically similar yet both deserve inclusion because they provide independent evidence. An evidence portfolio must therefore model task-specific roles and dependencies in addition to semantic redundancy.

Recent work moves toward set-level and reasoning-aware retrieval. DF-RAG diversifies factual perspectives to reduce redundant evidence and reports downstream gains rather than treating diversity as an end in itself [27]. NEST retains evidence paths that would be eliminated by locally optimal retrieval decisions, improving recall for complex reasoning [28]. COSMOS optimizes the utility of a coherent evidence subgraph under constraints [29]. User-centric evidence ranking evaluates how complementary evidence affects human verification effort [30]. RARE focuses on recall under redundant multi-hop conditions and makes visible the failure to recover all necessary facts [31]. These systems differ in task and mechanism, but collectively they support the premise that the value of evidence is conditional on what else has been retrieved.

Evidence Portfolio Retrieval extends this line of work by making the output contract explicit. The output is a bounded portfolio accompanied by a statement of the informational roles it attempts to cover, the source dependencies detected, the gaps that remain, and the rationale for including each item. This contract is useful even when the selector is implemented with conventional models. It changes what is measured, what is exposed to users, and which failures can be diagnosed.

The word portfolio is deliberate. In finance, the risk and utility of an asset cannot be judged independently of the holdings around it. In experimental design, the value of a measurement depends on what uncertainty remains. In software testing, a suite is evaluated by coverage and fault detection rather than by the quality of each test in isolation. Scientific evidence has the same set-level character. A portfolio can be redundant, unbalanced, dependent on one source lineage, or blind to a decisive counterexample even when every included document appears relevant.

The analogy should not be overstretched. Scientific evidence is not a collection of fungible assets, and epistemic roles are not stable numerical factors. The value of a document can change after reading it. A paper initially retrieved as supporting evidence may contain a limitation that turns it into a boundary case. A citation relationship may reveal that two apparently independent results share the same dataset. Portfolio construction is therefore iterative: initial role predictions guide selection, close reading revises the role assignments, and the revised gaps trigger new searches.

A useful distinction is between candidate generation and portfolio construction. Candidate generation should be broad and recall-oriented. It can use BM25, dense embeddings, late interaction, citation expansion, graph traversal, metadata filters, and domain-specific search APIs. Portfolio construction begins after a sufficiently rich pool exists. It assigns or predicts roles, estimates credibility and dependence, identifies redundant coverage, and selects a subset under a fixed budget. Combining the two stages obscures error attribution because a portfolio selector cannot recover evidence that the candidate generator did not include.

The fixed budget matters because downstream systems do not reason over unlimited corpora. Human reviewers have limited attention. LLMs have context, latency, and cost constraints even when context windows are large. More evidence can reduce accuracy through distraction, conflict, and positional effects [15]. A portfolio is therefore not an invitation to retrieve everything. It is a mechanism for allocating scarce attention to a set whose collective structure is appropriate for the decision.

The relevant comparison is consequently not between “RAG” and “EPR” as mutually exclusive products. RAG is the infrastructure that discovers and reads candidate evidence. EPR is a retrieval objective and control layer that can sit within an advanced RAG system. A conventional RAG path can answer direct questions, while the EPR path is invoked for decisions that require distributed, contrasting, or role-structured evidence. The system should route between them rather than impose portfolio overhead on every query.

This distinction also clarifies the research claim. EPR does not assert that relevance ranking is obsolete. It proposes that, for a definable class of scientific tasks, individual relevance is not a sufficient optimization target. At equal context and cost budgets, a downstream evaluator given an EPR portfolio should make better novelty, prior-art,

review, or hypothesis judgements than the same evaluator given the output of competitive top-k, reranked, and diversified RAG baselines. If no improvement is observed, the additional role structure should be removed or restricted to the tasks for which it is demonstrably useful.

The remainder of the book treats EPR under this standard. It is presented as an engineering hypothesis informed by modern retrieval research and by the needs of scientific automation, not as a proven replacement for RAG. The important shift is conceptual but testable: from retrieving documents that resemble a request to retrieving a body of evidence sufficient for a decision.

Chapter 8. Evidence Portfolio Retrieval: Definition and Scope

Evidence Portfolio Retrieval can be defined operationally as the selection, under a bounded attention budget, of a set of evidence units whose collective properties support a specified downstream epistemic task. The definition contains four commitments. The output is a set rather than an independently ranked list. The budget is explicit. The selection criteria depend on the downstream task. The portfolio is evaluated by what a reasoner or researcher can decide from it, not only by the similarity of its members to the original query.

An evidence unit can be a passage, claim, table, figure, experimental result, patent claim, software artefact, dataset record, or complete source. The appropriate granularity depends on the task. A novelty analysis may need both a patent-level view and claim-element passages. A peer-review system may need a figure together with its caption, method section, and sample description. A systematic review may treat a study as the unit for inclusion decisions but use outcome-level units during synthesis. EPR should not assume that every item in the portfolio has the same granularity.

Each selected item needs more than a relevance score. A minimal portfolio record identifies the source, exact evidence span, retrieval route, predicted role, covered facets, temporal status, provenance, confidence, and known dependencies on other items. It also records whether the evidence was directly observed, inferred from metadata, or generated as a search hypothesis. This distinction prevents the selection rationale from being confused with the scientific evidence itself.

The role vocabulary is task-specific. In novelty assessment, useful roles include holistic precedent, component precedent, combination precedent, structural analogue, enabling technology, independent rediscovery, boundary case, contradiction, and unresolved gap. In peer review, roles include support, replication, failed replication, methodological critique, alternative explanation, benchmark limitation, population boundary, leakage risk, and missing comparison. In scientific discovery, roles include mechanism, observation, anomaly, causal bridge, confounder, intervention, discriminating measurement, negative evidence, and transferable analogy. A universal role ontology would become vague. The practical design is a small common core combined with domain and workflow extensions.

A common core can still enforce important distinctions. Support and contradiction are directional relations to a target claim, not intrinsic document labels. Independence is relational: two papers may appear separate but share data, authors, protocols, or a review source. Directness distinguishes primary evidence from commentary. Scope records the population, conditions, and time period in which a result applies. Status records whether the source is current, superseded, corrected, retracted, or contested. These fields make the portfolio auditable and allow downstream agents to reason about more than topical similarity.

EPR begins with a task contract. The system must know what decision is being prepared, the evidence budget, corpus scope, time cut-off, admissible source types, required citation granularity, and acceptable uncertainty. “Find literature about this idea” is too underspecified. “Prepare up to twelve evidence units to assess whether the proposed mechanism was described before 1 January 2025, including direct precedents, component precedents, alternative terminology, and evidence that could invalidate the novelty claim” is an EPR task.

The next stage creates a task model. An LLM, semantic parser, or rule-based component decomposes the request into candidate facets, relations, claims, constraints, and evidence roles. This model is provisional. It is not accepted as scientific truth. Its function is to generate search routes and selection criteria. The system should expose the decomposition because an omitted facet can cause systematic retrieval failure. A user must be able to correct “privacy” into “differential privacy” or add a missing temporal constraint before expensive search proceeds.

Candidate generation should be deliberately plural. A lexical route preserves rare names, formulae, identifiers, and legal language. A dense route captures paraphrase and semantic similarity. Late interaction preserves local matches. Citation and reference expansion finds intellectual lineages. Metadata search enforces date, venue, source type, and status. Graph traversal follows claims, authors, methods, datasets, or patent families. Query expansion explores synonyms and historical terminology. An agentic route can issue new queries when early evidence changes the task model. EPR is intended to reuse these mature techniques and add a set-selection and evidence-accounting layer after candidate generation.

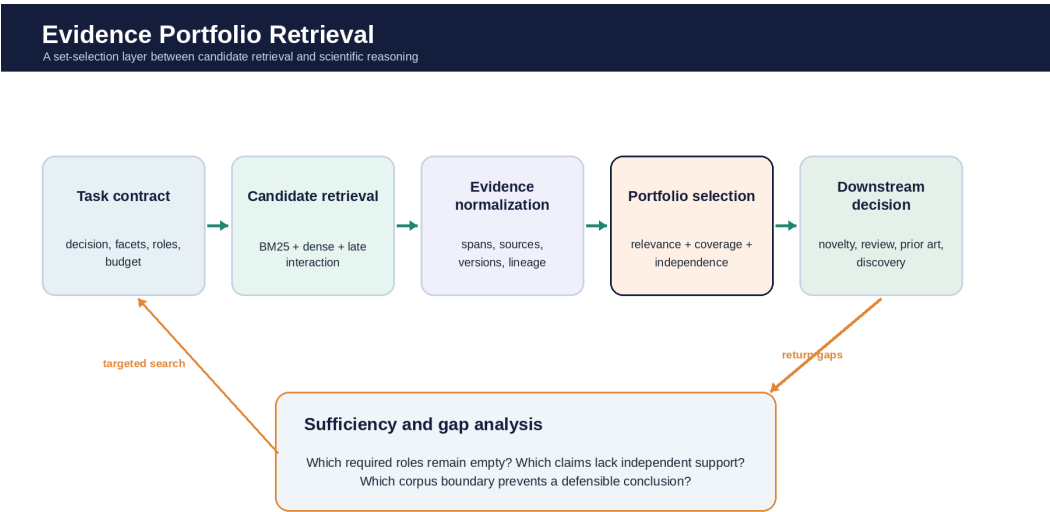


Figure 3. EPR reuses conventional candidate retrieval and adds an explicit set-selection, sufficiency, and gap-directed search layer before reasoning.

The candidate pool is then normalized. Duplicate versions are grouped. Retractions and corrections are attached. Passages are linked to their source and document

structure. Citation families and shared datasets are estimated. Exact dates and provenance are preserved. This stage is often neglected because it does not improve a retrieval benchmark immediately, yet it determines whether the portfolio later mistakes one study repeated across reviews for several independent confirmations.

Role assignment follows. A classifier can predict that a passage is a direct precedent, a methodological limitation, or an analogue. The prediction should be multi-label and calibrated because a source can play several roles. Role assignment can use the target claim, local passage, document metadata, citation context, and relationships to other candidates. A generative model may provide a rationale, but the stored record should contain normalized fields that can be inspected and evaluated. Role labels that cannot be supported by a source span should remain hypotheses.

Facet coverage is related but distinct. A candidate may cover human oversight and provenance while playing the role of a component precedent. Another may cover the same facets as a counterexample. Coverage answers what part of the target is addressed; role answers how the evidence participates in the judgement. Collapsing them loses important structure.

Credibility and suitability are also separate from relevance. A withdrawn preprint can be highly relevant but unsuitable as positive evidence. A review can be useful for navigation while inappropriate as the primary source for a contested claim. A patent can be decisive for prior art but weak evidence of empirical feasibility. EPR should not compute one universal quality score. It should represent source type, review status, methods, sample, replication, venue, conflicts, and provenance so that the downstream workflow can apply the appropriate policy.

Dependence estimation is one of the most important and difficult stages. Textual dissimilarity does not imply evidential independence. Two papers may use the same dataset, benchmark, laboratory, codebase, or derived result. Several articles may cite one original experiment without adding new evidence. A portfolio that ignores dependence can overstate support. Candidate graphs can record explicit citations, shared authors, dataset identifiers, trial registrations, patent families, and method lineage. Learned models may infer additional dependence, but uncertain links should not be treated as facts.

Selection converts the candidate pool into a bounded set. The selector balances direct relevance, facet coverage, role coverage, credibility, temporal validity, source independence, and redundancy. Hard constraints can require at least one direct precedent or one contradictory result when suitable candidates exist. Soft objectives can prefer multiple independent lineages. A relevance floor prevents diversity objectives from filling the portfolio with peripheral material. When the constraints cannot be satisfied, the system should report an explicit coverage deficiency and identify the unmet requirements.

The first selected set is rarely final. Close reading can reveal new terminology, a hidden assumption, a dataset lineage, or a gap. EPR therefore supports residual-directed retrieval. Here, the residual is the explicit difference between the evidence needs in the task model and the credible coverage provided by the current portfolio. If no source covers the relation between provenance and novelty scoring, the system formulates targeted searches for that relation, its synonyms, and plausible analogues. The loop stops

when coverage criteria are met, the budget is exhausted, or further search yields no material gain.

The primary output is a portfolio containing the selected evidence, role and facet map, dependency notes, unresolved conflicts, uncovered needs, and search log. A synthesis can be added downstream of the evidence package. This separation enables a human reviewer, another model, or a deterministic reasoner to inspect the same material. It also allows the portfolio to be re-evaluated when a source is retracted or a task policy changes.

The distinction between EPR and ordinary diversified retrieval can now be stated precisely. MMR selects items that are relevant to the query and dissimilar from what is already selected [23]. EPR may use MMR, but it also asks whether the set contains the required evidence roles, whether the sources are independent, whether a contradiction has been actively sought, whether temporal constraints are satisfied, and whether the portfolio is sufficient for the downstream decision. A non-duplicate set is not necessarily an epistemically adequate set.

The distinction from GraphRAG is equally important. A graph can organize entities and relations and can support multi-hop retrieval. An evidence portfolio may use graph retrieval, but useful members need not form one coherent subgraph. A counterexample from another domain can be intentionally disconnected. Conversely, a connected subgraph can consist entirely of one citation lineage and provide weak independent support. Connectivity is one feature, not the objective.

The distinction from agentic RAG is one of contract rather than implementation. An agent can execute EPR by decomposing the task, issuing searches, classifying roles, and filling gaps. Agentic behaviour alone does not ensure portfolio quality. Without explicit evidence requirements and stopping conditions, the agent may repeatedly retrieve similar sources or stop after finding confirming material. EPR supplies the state representation and evaluation target for the agentic loop.

The distinction from knowledge-graph reasoning is also sharp. EPR retrieves evidence for later reasoning. It may represent claims and dependencies, but it does not assume that all scientific judgement can be reduced to graph traversal or formal rules. Methods, measurements, uncertainty, and source credibility remain partly textual and contextual. The portfolio can feed a symbolic reasoner where appropriate while preserving the primary evidence for aspects that resist formalization.

EPR is therefore best understood as an intermediate system boundary. Upstream, heterogeneous retrievers discover candidates. Downstream, humans and agents evaluate hypotheses, write reviews, compare claims, or design experiments. EPR transforms an unstructured candidate pool into a bounded, role-aware, provenance-rich body of evidence. Its scientific value depends on whether this transformation improves decisions at a fixed cost.

That value will vary by task. Direct factual lookup should normally bypass EPR. Broad exploratory search may use a lightweight portfolio to expose themes but should not impose rigid roles. High-stakes research assessment benefits most when relevant evidence is distributed, contradictory, temporally constrained, or dependent on several

intellectual lineages. Routing is therefore part of the architecture: the system must recognize when a portfolio is warranted and when ordinary RAG is sufficient.

The concept is compatible with existing retrieval infrastructure. An organization can add EPR after a hybrid retriever and reranker, begin with manually defined roles and greedy selection, and later learn role prediction and portfolio policies from expert feedback. This incremental path allows each additional component to be evaluated against a well-engineered RAG baseline before it becomes part of the deployed architecture.

Chapter 9. Architectures and Design Patterns for Evidence Portfolios

A credible EPR implementation begins with ordinary retrieval engineering. The portfolio layer cannot compensate for a weak corpus, broken document parsing, poor candidate recall, or missing metadata. The design patterns in this chapter assume that the system can already search the relevant collections and localize evidence with reasonable accuracy. They address how to convert that capability into reasoning-ready evidence while preserving auditability and cost control.

The first pattern is broad-candidate, narrow-portfolio. Candidate retrieval uses a larger budget than the final context. It may collect hundreds of passages through several channels, but only a small subset is sent to expensive reasoning. This separation allows recall-oriented retrieval without overwhelming the downstream model. It also creates a stable experimental interface: different selectors can operate over the same frozen candidate pool, making causal comparison possible.

The candidate pool should preserve channel identity. Lexical matches, dense matches, late-interaction matches, citation neighbours, and agent-generated searches should not be merged into an anonymous list too early. Channel provenance helps diagnose why an item was found and supports calibrated fusion. Exact identifiers may deserve a lexical boost, while analogical candidates may arrive only through semantic or graph routes. Reciprocal Rank Fusion provides a robust baseline because it combines rankings without requiring directly comparable scores. Learned fusion can improve performance but needs domain-specific judgements and can obscure failure analysis.

The second pattern is explicit query decomposition with revision. The initial request is transformed into facets, relations, constraints, and evidence roles. For a novelty task, the facets may correspond to components of the proposed system, while the roles describe the kinds of prior art required. For a review task, the decomposition can include each major claim, method, dataset, baseline, and asserted contribution. The representation must be editable. A model-generated decomposition is a search plan, not an authoritative interpretation.

Decomposition should preserve uncertainty. If “model alignment” could mean value alignment, geometric alignment, or cross-modal alignment, the system should represent alternatives rather than commit silently. Candidate retrieval can explore each interpretation within a limited budget. Evidence and user feedback can collapse the ambiguity later. Premature disambiguation is a common cause of high-confidence retrieval failure.

The third pattern is role-conditioned search. Instead of issuing one expanded query, the system generates searches for specific evidence functions. A direct-precedent query emphasizes the complete mechanism. A component-precedent query isolates one element. A contradiction query includes failure, limitation, negative-result, and critique language but should not rely solely on those words because negative evidence is often described neutrally. An analogue query abstracts away the application domain while

preserving the functional structure. A boundary query searches for systems that approach but do not cross the claimed contribution.

Role-conditioned search can be implemented with templates, learned query generation, or an agent. The important requirement is traceability. Each query records which evidence need it was intended to satisfy. This makes it possible to determine whether an empty role resulted from corpus absence, poor query construction, or selector error.

The fourth pattern is passage-first evidence localization with source-aware promotion. Retrieval operates on passages because evidence is local, but selection reasons about their source. Several passages from one paper may be necessary to interpret a result, yet they should not be counted as independent evidence. A parent-child index retrieves fine-grained passages and promotes them with enough surrounding context. The portfolio groups related passages into evidence bundles tied to one source and one claim.

Evidence bundles solve several practical problems. They keep a result with its population, method, and limitation. They prevent an isolated sentence from being interpreted outside its conditions. They allow a table and accompanying explanation to travel together. They also make token budgeting more realistic: the unit admitted to the downstream reasoner is a coherent bundle, not an arbitrary chunk.

The fifth pattern is version-family normalization. Preprints, conference papers, journal versions, author manuscripts, corrections, and retractions are grouped. The selector chooses the appropriate version and records the family. Multiple versions can be shown when changes are themselves relevant, but they should not inflate support. Patent families require analogous treatment. Version handling is especially important in fast-moving AI research, where the same work may appear in several repositories and titles.

The sixth pattern is evidence-lineage modelling. The system builds a lightweight dependency graph from explicit and inferred signals. Citation edges are the simplest signal. Shared datasets, benchmarks, code repositories, trial registrations, cohorts, authors, institutions, and method templates provide additional links. The graph is not assumed to be complete. It supports warnings such as “three selected papers derive their central result from the same dataset” and selection preferences such as “include at least two independent empirical lineages when available.”

Lineage modelling should not become guilt by association. Shared authors do not automatically make results dependent, and citation does not imply evidential inheritance. The system should retain the evidence for each dependency assertion and use graded confidence. The human-facing language must distinguish a confirmed shared dataset from a possible intellectual connection.

The seventh pattern is relevance gating before diversification. Every candidate must pass a minimum task-relevance threshold or satisfy a justified analogue rule. This prevents the selector from adding unrelated material merely because it is different. The threshold can be role-specific. A direct precedent requires strong correspondence; a structural analogue may be farther in topic space but must match a stated relational pattern. Candidates below the gate can still seed exploratory searches, but they should not enter the evidence portfolio as if they supported the decision.

The eighth pattern is constrained coverage selection. The portfolio is assembled incrementally. At each step the selector estimates the marginal contribution of a candidate given the current set. Contribution can include uncovered facets, missing roles, an independent source lineage, a contradictory result, or a required date range. Redundancy and cost reduce the gain. A greedy algorithm is often the right starting point because it is transparent and fast. More complex learned policies should be justified only after the objective and labels are stable.

Coverage constraints can be hard or soft. A patent search may require coverage of every claim element before the portfolio is considered complete. A peer-review portfolio may prefer, but not require, contradictory evidence because none may exist. Hard constraints that cannot be met should produce a gap report rather than a low-quality substitute. Soft constraints should expose their degree of satisfaction.

The ninth pattern is deliberate contradiction search. Ordinary relevance retrieval tends to find material framed similarly to the query. A portfolio workflow issues independent searches for disconfirming evidence, failed replications, null results, retractions, limitations, and alternative mechanisms. The search is not a token gesture. A role remains unsatisfied until a credible candidate is found or the system reports that the searched corpus produced none under the recorded strategy.

Contradiction must be assessed at claim level. Two papers can appear opposed while studying different populations or endpoints. A source can contradict one claim and support another. The portfolio should therefore represent the target claim, polarity, scope, and conditions. LLM classification can assist, but decisive contradictions should be verified against source spans and, when possible, structured outcome data.

The tenth pattern is residual-directed iteration. After an initial portfolio is assembled, the system computes a plain-language gap model. Which facets are weakly covered? Which roles have no candidate? Which conclusions depend on one source family? Which relationship in the proposed idea has no precedent? Which claim lacks negative evidence? The gap model generates the next round of searches. This process is more disciplined than generic agentic browsing because each action corresponds to an explicit unsatisfied requirement.

The residual loop needs stopping rules. It stops when mandatory roles are covered, marginal gains fall below a threshold, the search budget is exhausted, or repeated searches fail to produce new credible evidence. The system records whether it stopped because the portfolio appears sufficient or because resources ran out. This difference is central to scientific honesty.

The eleventh pattern is portfolio compression without source loss. A downstream model may not need every full passage simultaneously. The system can create structured summaries of evidence bundles, but each summary sentence remains linked to source spans. Compression should preserve claims, conditions, uncertainty, and contradictions. It should not merge independent sources into an unattributed consensus statement. The original evidence remains available for inspection and targeted reopening.

Hierarchical context is useful here. The reasoner first sees a portfolio map containing roles, claims, and concise evidence statements. It opens detailed passages when making or checking a conclusion. This reduces distractor exposure while keeping the evidence

accessible. The design resembles retrieval over a case file rather than concatenation of search snippets.

The twelfth pattern is dual-output reasoning. The downstream agent produces both a substantive conclusion and an evidence-accounting report. The conclusion addresses the scientific task. The report states which portfolio items support each major claim, where evidence conflicts, which roles remain empty, and what uncertainty is caused by corpus or retrieval limitations. The accounting output makes it possible to evaluate whether a correct-looking answer was reached for the right evidential reasons.

The thirteenth pattern is claim-level citation enforcement. Every externally checkable claim in a synthesis should point to one or more portfolio evidence units. Citation validators test entailment at the cited span, not merely topical similarity. Unsupported claims can be removed, weakened, or sent back for additional retrieval. Scientific citation-faithfulness studies show that this verification remains necessary even in advanced agentic synthesis systems [45].

The fourteenth pattern is provenance-preserving human correction. A researcher can reclassify a source, reject an analogue, split a facet, mark two studies as dependent, or add a missing role. Corrections are stored as versioned annotations rather than silently modifying the corpus. They can improve future selectors, but the system records who made the change and for which task. A correction to one portfolio should not automatically become a universal scientific fact.

The fifteenth pattern is deterministic selection around probabilistic components. Encoders and LLMs can propose facets, roles, and scores. The final portfolio policy can apply explicit constraints and budgets reproducibly. Given the same candidate pool, model outputs, policy, and seed, selection should be replayable. This boundary allows model upgrades without losing the ability to audit past analyses.

The sixteenth pattern is policy-aware routing. Not every question enters the full EPR path. A router classifies direct lookup, broad exploration, synthesis, multi-hop verification, novelty assessment, prior-art search, review preparation, and discovery support. Direct lookup uses ordinary RAG. Complex assessment invokes role decomposition and portfolio selection. Uncertain routing can offer both a concise answer and an optional evidence dossier. This prevents EPR from becoming expensive ceremony.

The seventeenth pattern is portfolio caching by evidence need, not only by query string. Different users may ask semantically related questions that require similar evidence roles. A validated portfolio for a specific claim, time cut-off, and corpus snapshot can be reused and incrementally updated. Cache keys include task type, scope, date, and policy. Stale portfolios are invalidated when sources are retracted, new literature enters the corpus, or the decision contract changes.

The eighteenth pattern is explicit corpus and search disclosure. Every portfolio states which databases, repositories, languages, date ranges, source types, and access restrictions were included. This disclosure is indispensable for novelty and systematic review. A high-quality portfolio over an incomplete corpus is still incomplete. The system must not translate retrieval confidence into global novelty confidence.

The nineteenth pattern is adversarial separation. Retrieved content is untrusted data. Documents cannot issue tool commands, change selection policies, or instruct the agent

to ignore other sources. Prompt injection defenses separate source text from control instructions, sanitize metadata, and restrict tool permissions. Portfolio selection also creates a new attack surface: an adversary could publish many variations of one claim to occupy role slots or simulate independent support. Lineage and duplicate detection therefore become security controls as well as quality controls [52–54].

The twentieth pattern is benchmarkable modularity. Candidate generation, role prediction, dependence detection, selection, synthesis, and verification expose separate interfaces and logs. Each can be evaluated independently and in the full pipeline. This modularity is not bureaucratic architecture. It is how the programme distinguishes a real EPR gain from an improvement caused by a better encoder, a larger candidate pool, or a more capable generator.

A minimal implementation can be much simpler than the complete pattern catalogue. It can use hybrid retrieval, an LLM-generated facet and role plan, a cross-encoder for candidate relevance, a greedy set-cover selector with an MMR redundancy term, source-family deduplication, and a portfolio report with unresolved gaps. This is sufficient for a serious baseline. The system should become more complex only where benchmark errors demonstrate a need.

The resulting engineering discipline is selective. EPR does not require an ontology, graph, agent, or optimizer by default. Each component should correspond to a measured failure class, improve downstream decisions under an equal budget, survive ablation, and retain an understandable failure mode. The design patterns provide implementation options; empirical evidence determines which options belong in a deployed system.

Chapter 10. Evidence Portfolio

Retrieval in Scientific Automation

Scientific automation is often described as a sequence in which an AI reads papers, proposes ideas, runs analyses, and writes results. That description understates the information problem. At every stage, the system must decide which evidence deserves attention, which sources are independent, what has already been tried, what contradicts the current hypothesis, and what remains unknown. A model that generates sophisticated reasoning over a narrow or redundant evidence set can be less useful than a simpler model supplied with the right portfolio. EPR is therefore positioned in this programme as evidence infrastructure for research agents rather than as another conversational interface.

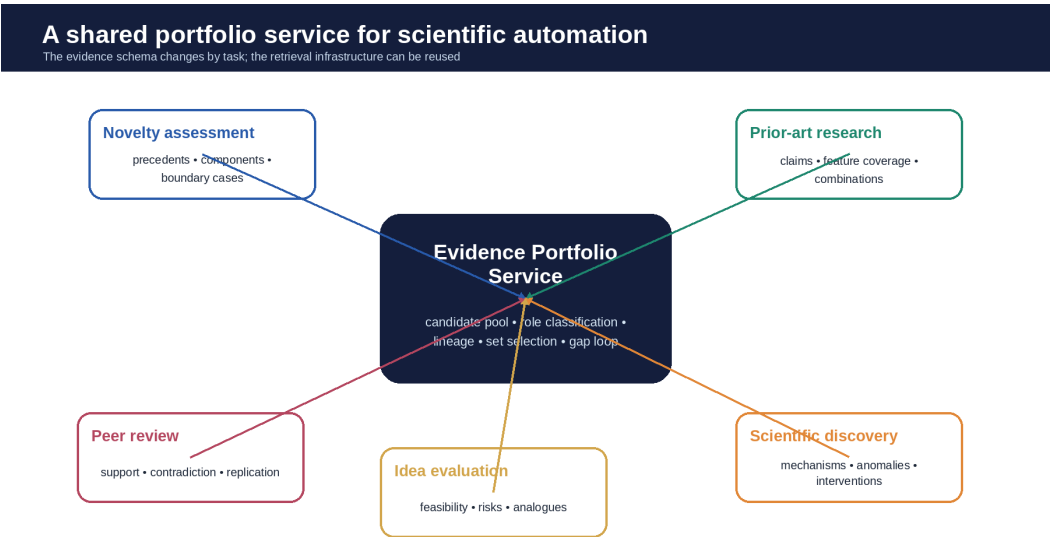


Figure 4. A shared evidence-portfolio service can support several scientific agents while keeping their evidence requirements distinct.

Novelty assessment is the most direct use case. A candidate idea can be decomposed into its purpose, mechanism, components, relations, assumptions, and claimed difference. The portfolio then seeks several distinct forms of precedent. A holistic precedent may describe nearly the same system. Component precedents show that individual elements are known. Combination precedents show that elements have already been assembled. Structural analogues reveal the same functional arrangement under different vocabulary or in another field. Boundary cases identify systems that approach the contribution but stop short of it. Independent rediscoveries show whether the idea emerged through separate lineages. Negative or contradictory evidence may show that the claimed novelty depends on a distinction without practical consequence.

The output should not be a universal novelty percentage. RINoBench demonstrates the difficulty of aligning automated novelty judgements with expert assessments even when generated rationales appear plausible [32]. NovBench similarly treats novelty assessment as a structured evaluation problem rather than a simple similarity score [33]. EPR contributes by improving the evidence basis: it can state that no holistic precedent was found within the searched corpus, that four components are well established, that one relation is weakly covered, and that the most relevant boundary case differs in a specified way. A human or specialized evaluator then judges whether that difference is scientifically meaningful.

An iterative novelty workflow begins with a preliminary portfolio and a residual description. Suppose a proposal combines automated claim extraction, provenance tracking, and counterfactual removal of cited evidence to estimate conclusion fragility. The first search may find strong precedents for the components but none for the complete mechanism. The residual is stated in language: no retrieved source propagates source removal through claim dependencies to quantify fragility. The next search targets this relation through alternative terminology such as sensitivity analysis, dependency cut, evidence ablation, argument graph robustness, and truth-maintenance systems. Novelty confidence should decrease as coverage expands and should remain conditional on corpus scope and date.

Prior-art research has similar structure but stricter temporal and legal requirements. A claim is decomposed into elements and relations. Retrieval searches patents, papers, standards, products, manuals, and archived materials using contemporary and historical terminology. The portfolio distinguishes complete references, element-level references, plausible combinations, analogous mechanisms, and documents that establish enabling knowledge. PatentEB provides a modern benchmark family for patent embeddings and retrieval [34], but a practical EPR evaluation must go beyond nearest-document performance to claim-element coverage and the recall of references that professional searchers consider material.

EPR does not make legal conclusions about novelty or inventive step. It supplies a reviewable search dossier. The dossier records filing and publication dates, patent families, jurisdictions, source versions, query history, and claim-element mappings. It can identify when several apparently distinct results belong to one family. It can also expose a dangerous condition: the system found every element separately but no admissible basis for combining the references. That condition is analytically different from finding a single anticipatory reference and should not be hidden behind one score.

Peer-review automation benefits from portfolios because manuscripts contain multiple claims and failure modes. A review agent can build one portfolio per major contribution or a shared portfolio with claim-level mappings. For a performance claim, the required evidence may include the closest methods, results on the same benchmark, independent replications, negative findings, known leakage, alternative explanations, and accepted evaluation protocols. For a methodological novelty claim, the portfolio resembles prior-art research. For an empirical generalization claim, it must represent populations, settings, and boundary conditions.

Tree-structured review systems such as TreeReview organize critique as a hierarchy of questions [57]. EPR can supply the evidence behind those questions. Instead of asking

a language model to critique from the manuscript and general memory, the system retrieves a portfolio for each high-impact issue. The reviewer sees the manuscript claim, supporting and conflicting evidence, source dependencies, and unresolved gaps. This architecture does not automate final judgement; it makes automated assistance less dependent on model recall and less vulnerable to the first plausible related papers retrieved.

Automated reviewers have been shown to miss faulty reasoning even when they produce fluent and superficially useful comments [58]. EPR addresses one part of that failure: missing external evidence and insufficient contrast. It cannot by itself detect an invalid proof, an inappropriate statistical model, or manipulated data. Those require specialized analyses. The portfolio should therefore connect to other agents, including statistical reviewers, reproducibility checkers, code executors, image-forensics tools, and policy checkers. EPR's role is to ensure that each agent can request the external evidence its analysis requires.

Systematic review and evidence synthesis are natural but demanding use cases. Conventional systematic review separates protocol, search, deduplication, screening, extraction, risk-of-bias assessment, and synthesis. A portfolio selector cannot replace the exhaustive-search requirements of a formal review. It can assist scoping, screening prioritization, evidence-map construction, and context preparation for synthesis. RAG4SR-CS-200 represents this workflow with protocol-oriented artefacts rather than isolated questions [35]. IntraBench and PaperScope similarly move evaluation toward literature-review and multi-document scientific tasks [36–37].

For rapid or living reviews, EPR can maintain portfolios organized by intervention, population, outcome, methodology, and evidential role. New studies are compared with the current portfolio not only for relevance but for whether they change a conclusion, add an independent lineage, cover an underrepresented population, or contradict prior findings. The system can flag when a portfolio is saturated with one study design or one geographic region. It can also identify where a concise synthesis is unsafe because coverage remains sparse or conflicting.

OpenScholar demonstrates the potential of retrieval-intensive scientific synthesis over a very large open-access corpus, while PaperQA2 shows how iterative search and evidence-grounded answering can outperform less structured approaches on literature questions [21–22]. EPR can complement these systems by controlling the evidence set before synthesis. Instead of relying on the generator to notice redundancy and missing perspectives inside a long context, the portfolio layer makes those properties explicit and measurable.

Scientific discovery requires a different portfolio schema. A hypothesis should be surrounded by evidence for the phenomenon, candidate mechanisms, causal bridges, anomalies, negative results, confounders, interventions, and measurements capable of distinguishing explanations. A portfolio can also include transferable mechanisms from other domains, but analogies must state the preserved structure and the broken assumptions. The objective is not to maximize novelty. It is to expose a hypothesis to the evidence most likely to confirm, refine, or kill it.

Recent systems such as AI co-scientist frameworks and multi-agent scientific-discovery systems illustrate the move from question answering toward

hypothesis generation, debate, experimental planning, and iterative refinement. ScienceAgentBench evaluates whether agents can execute parts of scientific workflows rather than merely discuss them. These developments make evidence control more important, not less. An agent that searches only for support can generate impressive but fragile research programmes. An EPR controller can require disconfirming and boundary evidence before a hypothesis advances.

A discovery portfolio can be dynamic. Early in exploration, diversity and analogies receive more budget. As a mechanism becomes precise, direct empirical and methodological evidence dominates. Before an experiment, the portfolio emphasizes confounders, measurement validity, and prior negative results. After results arrive, it emphasizes alternative explanations and replication. The portfolio is therefore linked to the state of the research programme, not only to a static query.

Idea evaluation is a broader organizational use case. Research groups and venture studios receive many proposals that combine known components under new narratives. A portfolio can distinguish an idea with a close complete precedent from one whose components are known but whose integration is unusual. It can reveal an apparently original idea that is standard under different terminology in another field. It can also show that a technically novel mechanism addresses a problem with weak evidence of importance. Decision makers can then score opportunity, feasibility, scientific risk, and strategic relevance separately rather than treating novelty as a proxy for value.

Portfolio retrieval is useful here because evaluation requires balanced evidence. A persuasive proposal naturally supplies supporting arguments. The portfolio system independently searches for prior attempts, market or scientific substitutes, failure modes, enabling dependencies, and claims that would falsify the thesis. The result can feed an investment committee, research steering group, or grant-writing process. The portfolio is not an automatic decision; it is a compact adversarial dossier.

Research-integrity monitoring offers another application. A claim-centric portfolio can collect the original result, replications, corrections, retractions, image or data concerns, and downstream papers that depend on the claim. When source status changes, the system identifies portfolios and conclusions affected by the update. Dependency-aware evidence is particularly valuable because retraction effects do not propagate automatically through the literature. A portfolio can reveal that several apparent confirmations merely reused the original dataset or cited the same unverified measurement.

Contradiction detection belongs in this workflow but must be scoped carefully. Textual contradictions can be caused by different populations, definitions, endpoints, or time periods. EPR can retrieve and cluster apparently conflicting claims, align their conditions, and present the unresolved contrast. It should avoid declaring the science inconsistent until scope and methods have been compared. The portfolio's contribution is to make the competing evidence visible and structured enough for specialized reasoning.

Executable science provides the long-term architectural context. Scientific documents increasingly need machine-readable claims, methods, datasets, transformations, provenance, and dependencies. EPR can operate over free text today while producing structured annotations that make later retrieval and reasoning more precise. As scientific artefacts become executable or semantically compiled, evidence units can link directly to

workflows, code, data, and reproducible results. The portfolio can then include not only statements but executable checks.

For example, a peer-review portfolio may contain a claim from a manuscript, two prior studies, the dataset version used, a benchmark definition, an executable reproduction script, and a statistical critique. A discovery portfolio may include a mechanistic hypothesis, observations, a causal model, and candidate experiments represented as workflows. EPR does not execute these artefacts itself. It identifies and assembles the evidence package required by agents that can.

This integration supports a multi-agent research environment. A planning agent formulates evidence needs. Retrieval agents search heterogeneous sources. A portfolio agent selects complementary evidence. A credibility agent evaluates source suitability. A novelty agent compares the candidate idea with the portfolio. A peer-review agent applies manuscript-specific criteria. A compliance agent checks privacy, copyright, and regulation. A provenance service records every source, transformation, and decision. The portfolio is the shared boundary object that prevents agents from reasoning over incompatible or undocumented evidence subsets.

Human interaction remains essential. Researchers need to inspect why an item was included, replace weak sources, adjust roles, and request additional searches. A useful interface presents the portfolio as a map of the decision rather than a feed of documents. It shows which facets are covered, which sources are independent, where evidence conflicts, and what remains unsearched. It allows the user to open the exact passage and surrounding context. It also separates model suggestions from accepted annotations.

The application cases differ in stakes and evaluation. Novelty assessment emphasizes precedent coverage and calibrated uncertainty. Prior art emphasizes temporal admissibility and claim elements. Peer review emphasizes flaw detection and balanced evidence. Systematic review emphasizes protocol compliance and exhaustive accounting. Discovery emphasizes useful hypotheses and discriminating experiments. Idea evaluation emphasizes decision utility. A single generic portfolio template would flatten these differences. The research programme should therefore build a common platform with task-specific schemas, policies, and benchmarks.

The common value proposition is nevertheless coherent. Scientific automation needs more than answers. It needs evidence packages that can survive inspection, correction, reuse, and changing conclusions. RAG supplies access to external knowledge. EPR attempts to turn that access into bounded, role-aware, reasoning-ready evidence. Whether the attempt succeeds must be established separately in each use case, but the need for the capability is already visible across the emerging research-agent ecosystem.

Chapter 11. Formal Objectives and Optimization Methods

This chapter presents mathematical formulations that can clarify EPR and defines the limits of what those formulations establish. A compact objective function is useful for implementing and comparing selectors. It does not by itself establish that the selected evidence is epistemically sufficient, that a novelty judgement is valid, or that the encoded roles correspond to expert reasoning. Mathematics specifies an operational proxy; empirical evaluation determines whether the proxy serves the intended task.

Let a query or decision task be denoted by q , a candidate pool by D , a selected portfolio by S , and a fixed budget by B . Ordinary top- k retrieval approximately selects the B candidates with the largest individual relevance score $r(q,d)$. The selection of one candidate does not directly alter the score of another. In EPR, utility is set-dependent. The relevant quantity is the marginal contribution of candidate d given the portfolio already selected:

$$\Delta(d \mid S, q) = U(S \cup \{d\}, q) - U(S, q).$$

This notation captures the central idea without deciding how utility is measured. A near-duplicate may have high $r(q,d)$ but low marginal contribution. A candidate that covers an unmet role may have moderate direct relevance but high marginal contribution. The portfolio is selected to maximize U subject to a budget and admissibility constraints.

A simple implementable objective can combine direct relevance, facet coverage, role coverage, independence, and redundancy:

$$U(S,q) = \alpha R(S,q) + \beta F(S,q) + \gamma E(S,q) + \delta I(S) - \lambda C(S) - \mu K(S).$$

R measures relevance to the task. F measures covered facets. E measures covered epistemic roles. I rewards independent evidential lineages. C penalizes redundancy or dependence. K measures cost in tokens, documents, latency, or money. The coefficients do not have natural universal values. They represent a policy that must be tuned and audited for a task.

Coverage can be represented by weighted concepts or requirements. Suppose the task model contains needs n in a set N , each with importance w_n . Candidate d has a calibrated coverage value $c(d,n)$ between zero and one. A saturating set function is:

$$F(S,q) = \sum_{n \in N} w_n \max_{d \in S} c(d,n).$$

The maximum prevents repeated coverage of one need from adding unlimited value. Variants can require multiple independent sources for high-stakes needs. If $m_n(S)$ is the number of independent lineages covering n , a capped function such as $\min(m_n(S), t_n)$ can reward replication up to a desired target.

Role coverage has the same mathematical form but should not be confused with facets. The facet “human oversight” can be addressed by a supporting precedent, a

failure case, or a regulatory requirement. If P is the set of required roles and $p(d,j)$ is the probability that d fulfils role j , then a simple role utility is:

$$E(S,q) = \sum_{\{j \in P\}} v_j \max_{\{d \in S\}} p(d,j).$$

This formulation also identifies a calibration requirement. If p is produced by an uncalibrated LLM, the optimizer can convert model confidence into apparent certainty about role satisfaction. Role probabilities must therefore be evaluated, and decisive roles may require source-span verification before they count as satisfied.

Redundancy can be measured through pairwise similarity:

$$C_{pair}(S) = \sum_{\{d_i, d_j \in S, i < j\}} sim(d_i, d_j).$$

This is easy to compute but semantically crude. Two independent replications can be similar and valuable. Two textually different reviews can depend on the same primary experiment. A stronger penalty combines semantic overlap with lineage dependence. If $dep(d_i, d_j)$ estimates shared evidence origin, a policy can penalize dependence more strongly than wording similarity.

Maximal Marginal Relevance is a standard baseline for this family of decisions [23]. At each step it chooses the candidate maximizing a weighted combination of query relevance and dissimilarity from the selected set. MMR is simple, transparent, and often effective. It can be incorporated into EPR when similarity is supplemented by task roles and evidence dependencies. Any claimed EPR contribution must be measured beyond the gains already provided by MMR.

Intent-aware methods such as xQuAD explicitly model subtopics or intents [24]. Set-cover and submodular formulations reward diminishing returns for repeated coverage. Determinantal point processes assign higher probability to diverse sets through determinants [26]. Graph formulations optimize coverage and connectivity, as in COSMOS [29]. Learned selectors can optimize downstream reward. These mathematical families are available tools. The research question is which one best represents evidence needs and generalizes across tasks.

Submodularity is useful because it supports efficient greedy selection with approximation guarantees. A set function is submodular when marginal gains decrease as the selected set grows, and common coverage functions have this property. If U is monotone submodular under a cardinality constraint, greedy selection has a known approximation guarantee. This result supports algorithm design but does not validate the role labels or establish that the coverage objective captures scientific sufficiency. Optimization quality and objective validity must be evaluated separately.

Hard constraints can represent non-negotiable requirements. The portfolio may need at least one source before a temporal cut-off, one primary source, or coverage of every patent claim element. Mathematically, the selector maximizes U while satisfying constraints $g_i(S) \geq b_i$ and cost $K(S) \leq B$. In practice, constraints can conflict or be impossible. The system must return infeasibility and identify unsatisfied requirements rather than relax them silently.

Uncertainty can be propagated through expected utility, robust optimization, or scenario analysis. If role and relevance scores are uncertain, one can optimize the

expected portfolio or the worst case over plausible scores. These approaches may improve stability, but additional formal complexity is justified only after the underlying predictions are sufficiently accurate and calibrated. A simpler initial approach is to expose confidence intervals, avoid sharp thresholds near uncertainty boundaries, and request human review for high-impact selections.

Portfolio regret is a useful evaluation concept. Let S^* be an oracle portfolio under the same budget and U^* its downstream utility. The regret of selected portfolio S is $U^* - U(S)$. In real scientific tasks, the oracle is rarely known. Expert-constructed portfolios can approximate it, but experts may disagree and discover additional evidence during evaluation. Regret should therefore be treated as a benchmark construct, not an absolute measurement of epistemic loss.

Formal treatment of novelty requires particular care. A candidate idea can be represented by an embedding x and compared with a corpus. Distance to the nearest neighbour, reconstruction error from known prototypes, density estimates, one-class classifiers, or regional representations can measure how unusual x is relative to the indexed collection. These quantities measure corpus-conditioned coverage or atypicality; they are not direct measures of scientific novelty.

A residual can nevertheless be operationally useful. Suppose the system explains a candidate idea through known components and identifies one relation that lacks coverage. The residual is then a symbolic or textual gap: “no source in the current portfolio links evidence provenance to counterfactual novelty scoring.” That statement can direct further search and remains conditional on the corpus and search protocol. A single scalar novelty score is insufficient because it obscures the corpus boundary, temporal cut-off, representation error, and the distinction between a new concept and an out-of-domain input.

Geometric representations require the same empirical discipline. Concepts can be represented as points, clusters, subspaces, boxes, cones, or distributions. Non-negative least squares can reconstruct a candidate vector as a positive combination of known directions, while vector-symbolic architectures can bind roles and relations. These models can test whether a representation is compositionally explainable. Preliminary evaluations conducted within this programme did not show a robust external advantage for a conical or VSA core over simpler retrieval and coverage methods. The result does not exclude future geometric approaches, but it provides no current basis for making them a required component.

Regional geometry is therefore optional rather than architectural. A future geometric model can be added as a candidate channel if it improves role coverage, analogue retrieval, or novelty decomposition against competitive baselines. The same criterion applies to graph methods, symbolic logic, and multi-agent deliberation: each mechanism is retained only when its incremental contribution is measurable and attributable.

Bayesian formulations also require assumptions that are often difficult to justify in open scientific literature. Publication bias, dependent studies, heterogeneous methods, and selective reporting complicate the assignment of priors and likelihoods and violate simple independence assumptions. Formal evidence aggregation is appropriate where study designs and outcome models support it. In other settings, the system should

present structured evidence and uncertainty without implying a level of posterior precision that the available data cannot sustain.

Information-theoretic formulations can describe the value of a document as expected reduction in uncertainty. This is attractive for active search and experiment design. The difficulty is defining the state of uncertainty and obtaining calibrated probabilities. In discovery workflows, approximate information gain may still rank candidate searches or experiments. The system should validate that ranking through held-out decisions rather than treating entropy reduction as intrinsically meaningful.

The most defensible use of mathematics in EPR is comparative and operational. Formalization can specify budgets, constraints, scoring components, and ablations. It can make a selector reproducible and reveal trade-offs. It can support optimization once evidence roles and task metrics are empirically grounded. It cannot define novelty, credibility, importance, or scientific sufficiency independently of expert practice and validated task outcomes.

Every mathematical term should therefore correspond to an observable claim. If the objective contains role coverage, the benchmark must show that roles are identified with adequate reliability and improve downstream decisions. If it rewards independence, the system must detect source dependence accurately. If it claims lower redundancy, experts should judge the selected evidence less repetitive without losing critical replication. If it claims novelty sensitivity, it must succeed on dated, domain-covered, expert-labelled novelty cases. Equations specify hypotheses and implementation contracts; experiments determine whether they are useful.

The EPR programme uses formal models in this operational sense. It begins with transparent greedy baselines, evaluates each signal separately, and adopts more sophisticated optimization only when task data supports the additional complexity. This sequence is intended to produce a reliable system for scientific work while preserving clear attribution of any measured gain.

Chapter 12. Comparative Evaluation Against Modern Retrieval Baselines

EPR should be assessed as an empirical claim about evidence allocation. The central experiment can be stated directly: at the same evidence and computational budget, does a downstream scientific decision improve when the evaluator receives a role-aware portfolio rather than the output of a competitive RAG pipeline? The remaining evaluation design is intended to make the result attributable and reproducible.

Evaluation must follow the causal chain

A retrieval improvement is useful only if it survives into the scientific decision

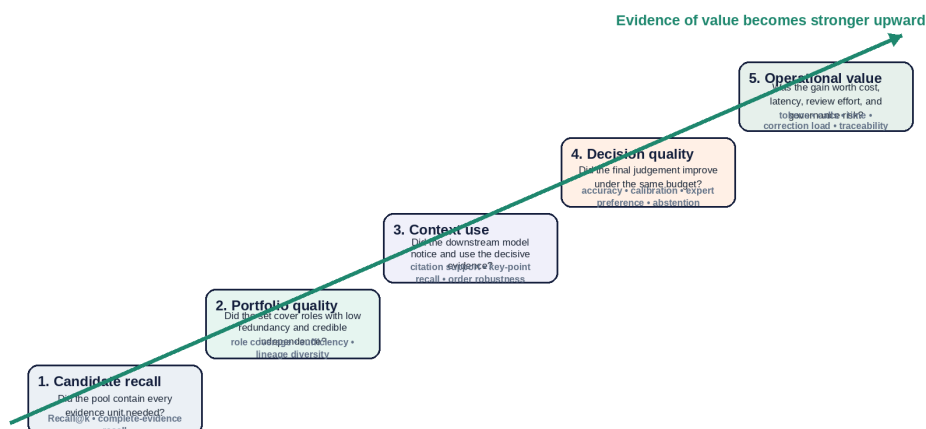


Figure 5. Evaluation must proceed from candidate recall to operational value. A gain at an early layer is not sufficient evidence of scientific utility.

The comparison must use competitive retrieval baselines. An evaluation limited to unreranked BM25 or single-vector cosine similarity would primarily measure established improvements in retrieval engineering rather than the contribution of EPR. The minimum baseline stack should include BM25, a domain-appropriate dense retriever, hybrid fusion, a late-interaction retriever such as ColBERTv2, and cross-encoder reranking. The baseline should deduplicate source versions and use the same candidate corpus, document parser, chunking policy, metadata filters, and context budget as EPR.

Diversification baselines are mandatory. MMR tests whether relevance plus redundancy control is sufficient [23]. xQuAD or another intent-aware method tests explicit facet coverage [24–25]. A learned diversification model, where training data permits, tests whether generic set selection absorbs the proposed advantage. DF-RAG, NEST-like multi-path retention, and graph/subgraph selectors provide stronger reasoning-aware comparisons [27–29]. An LLM selector given the same candidate pool should be included because it may infer epistemic roles without a specialized architecture.

The generator or downstream reasoner must be fixed across retrieval conditions. If EPR uses a stronger model, longer prompt, or more tool calls, the result is uninterpretable. The comparison should freeze the model version, decoding policy, task instructions, and citation verifier. The only intentional difference is the evidence package. Where possible, a non-generative evaluator should also consume the portfolio, reducing the chance that results depend on one model's prompt sensitivity.

The budget requires several definitions. Context budget controls total evidence tokens exposed to the reasoner. Document budget controls the number of distinct sources. Retrieval budget controls candidate-pool size and search calls. Compute budget controls embedding, reranking, and selection cost. Latency and monetary cost should be reported separately. A portfolio that improves quality by reading ten times more candidates may still be valuable for high-stakes review, but it should not be described as a free retrieval improvement.

Evaluation should begin with task instances whose evidence requirements are independently annotated. Experts define the target decision, relevant corpus cut-off, critical facets, useful evidence roles, source dependencies, and an acceptable range of conclusions. The system developers should not author all instances because they will unconsciously encode their own role schema into the benchmark. At least part of the test set should come from external cases and hidden annotations.

The first experimental layer evaluates candidate generation. It measures whether the union of retrieval channels contains all evidence that an expert portfolio requires. Recall is reported at several candidate-pool sizes. For distributed evidence, complete-evidence recall measures whether every required unit is present. For role-based tasks, role-eligible recall measures whether at least one credible candidate for each required role reaches the pool. This layer prevents a selector from being blamed for evidence it never received.

The second layer evaluates role, facet, and dependency predictions. Multi-label precision, recall, calibration, and source-span support are required. Role accuracy should be stratified by task and source type. Dependence detection should be tested on known dataset reuse, patent families, duplicate versions, citation derivation, and independent replications. A portfolio system that cannot identify these properties cannot safely optimize them.

The third layer evaluates selection as a set. Conventional nDCG, precision, and recall remain necessary because a portfolio should not achieve coverage by abandoning relevance. Set-level measures include facet coverage, role coverage, source-family coverage, complete-portfolio rate, redundancy, and independence. Coverage is counted only when a credible evidence unit supports the claimed facet or role. Model-generated labels cannot grade themselves.

Evidence sufficiency is the most important intermediate measure. An expert or blinded evaluator receives only the selected portfolio and task statement, not the full corpus. The evaluator reports whether the evidence is sufficient to make the target judgement, what is missing, and whether irrelevant material interferes. A portfolio can have excellent formal role coverage and still be insufficient because the selected evidence is weak, mis-scoped, or badly localized.

The fourth layer evaluates downstream decisions. Novelty assessment can measure agreement with expert novelty classes, identification of known components, recall of decisive precedents, calibration, and quality of uncertainty explanations. RINoBench and NovBench provide starting points, but the evaluation should include retrieval traces and fixed corpus snapshots so that failures can be attributed [32–33]. A system should receive credit for a correct abstention when corpus coverage is inadequate and be penalized for converting an empty search result into a novelty claim.

Prior-art evaluation should use claim-element mappings and material references rather than only document similarity. PatentEB can test patent retrieval channels [34]. A dedicated portfolio benchmark should measure complete claim-element coverage, recovery of references under alternative terminology, family deduplication, chronological validity, and whether a professional evaluator considers the dossier useful. Legal conclusions remain outside the automated metric.

Peer-review evaluation should test whether portfolios improve the detection of unsupported claims, missing baselines, flawed comparisons, contradictory literature, leakage, and overgeneralization. Review quality can be judged by domain experts under blinded retrieval conditions. The final prose should not be the only object of evaluation. The benchmark should record which evidence triggered each criticism and whether the criticism remains valid after source inspection.

Scientific-discovery evaluation is harder because the value of an evidence set can emerge over time. Near-term tests can measure hypothesis ranking against held-out discoveries, selection of discriminating experiments, identification of known failure mechanisms, and expert preference for research plans. ScienceAgentBench and recent co-scientist systems provide task structures, but EPR evaluation should isolate the retrieval contribution rather than compare whole agents with different capabilities.

Systematic-review evaluation can use screening recall, study-design coverage, risk-of-bias evidence, extraction accuracy, and synthesis correctness. The portfolio mode is most appropriate for scoping and bounded synthesis; formal reviews still require exhaustive accounting. RAG4SR-CS-200 and related corpora can test whether role-aware selection improves protocol-grounded outputs under a fixed context budget [35].

The evaluation should include direct questions on which EPR is expected to match or underperform ordinary retrieval. Exact lookup, single-hop factual questions, and one-document evidence tasks prevent the programme from optimizing only for distributed-evidence cases. A routing system is successful if it invokes ordinary RAG for these tasks and reserves EPR for decisions that require a structured body of evidence. End-to-end evaluation should therefore include routing accuracy and the cost of unnecessary portfolio construction.

Query difficulty must be characterized. Relevant dimensions include the number of facets, number of evidence roles, terminology mismatch, evidence distribution across sources, contradiction, temporal constraint, source dependence, corpus redundancy, and required analogical distance. EPR’s hypothesis predicts larger gains when several roles are distributed across a redundant corpus and the context budget is tight. Reporting only an overall average could hide both genuine specialization and failure elsewhere.

Corpus construction requires temporal discipline. Novelty and prior art must be evaluated against material available before a cut-off date. Retrieval models can be pretrained on later text, creating subtle leakage even when the indexed corpus is time-bounded. The book’s programme does not claim that all such leakage can be eliminated. It requires disclosure of model training dates where known, sensitivity analysis with lexical baselines, and caution in interpreting time-specific novelty results.

Human evaluation needs an explicit protocol. Evaluators should be blind to the retrieval method. Evidence order should be randomized where order is not part of the condition. Expertise, conflicts, and adjudication procedures should be reported. Inter-annotator agreement is not merely a quality score; disagreement can reveal that the task lacks a stable gold standard. For novelty and scientific importance, calibrated distributions or structured rationales may be more honest than forcing one label.

Downstream model evaluation also needs order controls. Long-context models are sensitive to where evidence appears [15], and RAG systems can change answers when retrieval order is permuted [51]. Each portfolio condition should be tested under multiple evidence orders or use a fixed deterministic ordering policy. Otherwise, a selector can appear better because it happens to place one decisive passage near a favoured context position.

The experiment should include a no-retrieval and full-context condition where feasible. The no-retrieval condition measures how much the model already knows or guesses. The full-context condition estimates the loss introduced by evidence selection, although it can itself suffer from overload. A human-curated oracle portfolio provides an approximate upper bound. These conditions locate the value of EPR within the larger system.

Ablation should be specified before final evaluation. Candidate generation, facet decomposition, role prediction, contradiction search, lineage penalties, portfolio optimization, and residual iteration are each removed in turn. A simple greedy coverage selector is compared with MMR, learned set selection, and LLM selection. The ablations identify which components account for any gain and whether a simpler configuration preserves most of the value.

Robustness tests should include paraphrased queries, shortened and expanded descriptions, multilingual terminology, noisy OCR, duplicated documents, citation spam, adversarially inserted instructions, and corpus updates. A novelty system should be tested with out-of-domain ideas to ensure that insufficient domain coverage triggers abstention rather than high novelty. A peer-review system should be tested with deliberately misleading abstracts whose methods sections reverse the apparent result.

Statistical analysis must match the paired design. Retrieval and downstream results can use paired bootstrap intervals and randomization tests. Effect sizes and quality-cost curves matter more than isolated p-values. A five-point improvement in flaw detection at equal budget is meaningful. A statistically significant 0.2-point gain that requires triple the cost may not be. Results should be reported for each domain and task regime.

The programme needs predeclared continuation criteria. A reasonable initial gate is that EPR improves the primary downstream decision by at least five absolute percentage points over the strongest conventional and diversified baseline in two distinct scientific domains, without losing more than two points on direct relevance and without more than

doubling normalized cost. The exact values can change before the study, but they should not be changed after seeing the results.

A second gate concerns attribution. At least one EPR-specific component—role coverage, evidence-lineage control, contradiction requirements, or residual-directed iteration—must contribute materially beyond generic MMR or an LLM reranker. If all gains come from broader candidate retrieval or a stronger generator, EPR is not the explanation. The appropriate product would be improved RAG.

A third gate concerns reliability. The system must identify when no adequate portfolio can be assembled. In high-stakes tasks, an explicit insufficiency state is more valuable than a superficially complete dossier. Calibration is evaluated by whether low-confidence or incomplete cases correspond to higher error. A system that gains average accuracy while remaining confidently wrong on corpus gaps is not suitable for scientific automation.

A fourth gate concerns user value. Domain experts should prefer the portfolios for real work, not only produce higher benchmark scores. Time-to-decision, number of additional searches, overlooked decisive evidence, and correction effort are practical outcomes. Evidence Ranking research suggests that complementary ordering can reduce verification effort [30], but EPR must demonstrate the effect in its target workflows.

The benchmark programme should release corpus snapshots where licensing permits, task contracts, role schemas, candidate pools, human annotations, selection traces, prompts, model versions, and cost records. This level of reproducibility is itself part of the scientific contribution. A system designed to improve research evidence should not be evaluated through opaque prompts and irreproducible web searches.

The most informative outcome would be a performance map rather than a universal ranking of systems: ordinary RAG may be preferable for direct lookup; late interaction for fine-grained passage ranking; EPR for distributed-evidence decisions under tight budgets; graph retrieval for relation-intensive tasks; and long-context processing for small, complete corpora. Such a map would define the conditions under which EPR is an appropriate engineering option and the conditions under which it is unnecessary.

A valid negative result would show that query decomposition, CoBERTv2, cross-encoder reranking, and MMR or an LLM selector already provide the same downstream benefit. In that case, the role vocabulary may remain useful for interfaces and evaluation, while a distinct EPR selector would not be warranted. The programme is designed to support that conclusion if the evidence leads to it.

Chapter 13. Governance, Security, Provenance, and Accountability

A retrieval system that influences novelty judgements, reviews, research priorities, or funding decisions becomes part of scientific governance. Its risks are not limited to wrong answers. It can shape which literatures are visible, which evidence is treated as independent, which disagreements appear important, and which ideas are labelled derivative or promising. EPR increases this influence because it selects a deliberately small portfolio intended to frame downstream reasoning. Governance must therefore be designed into the evidence pipeline.

The first principle is scope disclosure. Every portfolio identifies the corpus, repositories, languages, source types, access limitations, and temporal cut-off. A novelty report over open-access English papers is not a global prior-art search. A peer-review portfolio built from one publisher's index does not represent the whole field. The interface should place scope near the conclusion, not bury it in metadata.

The second principle is provenance at evidence-unit level. Each selected claim links to an exact passage, table, figure, dataset record, or executable artefact. The system records document version, retrieval time, transformation steps, and selector rationale. Generated summaries are visibly distinct from quoted or extracted evidence. Provenance enables verification and later impact analysis when a source changes.

The third principle is source-state awareness. Retractions, expressions of concern, corrections, superseded versions, and patent-family relations must propagate to portfolios. A retracted study can remain relevant as historical or negative evidence, but it cannot silently support a current conclusion. The system should retain the prior portfolio for audit while marking which conclusions require reassessment.

The fourth principle is epistemic status separation. Retrieved source claims, model interpretations, inferred relations, search hypotheses, and human decisions occupy different layers. A model may infer that a paper is a boundary case, but the role remains a classification until reviewed. An evidence unit may report an empirical result without making the result true. A portfolio may be complete relative to its schema while the schema omits an important question. The system should avoid interfaces that collapse these layers into one authoritative narrative.

The fifth principle is calibrated abstention. EPR must be able to state that candidate recall appears inadequate, a required role is empty, source independence cannot be established, or the corpus does not cover the domain. Abstention should be specific. "Insufficient coverage of pre-2010 non-English patents" is actionable; "low confidence" is not. The system can propose the next search without presenting the current absence as evidence of novelty.

The sixth principle is contestability. Users need to see why a source was selected and why another was excluded. They can change the evidence budget, temporal cut-off, required roles, relevance floor, and dependence policy. A decision influenced by the

portfolio should preserve the portfolio version and user corrections. This is especially important when novelty or review outputs affect careers, publication, patents, or investment.

The seventh principle is diversity of intellectual lineages without artificial balance. Portfolio selection should reduce accidental domination by one school, venue, language, or dataset. It should not force equal representation of positions when evidence quality differs dramatically. The policy should distinguish seeking disconfirming evidence from manufacturing a false controversy. Source credibility and methodological relevance remain part of selection.

Publication and accessibility bias require explicit monitoring. Highly cited and open-access documents are easier to retrieve and validate. Negative results, local journals, older terminology, supplementary materials, and non-English work may be underrepresented. Portfolio dashboards can report concentration by venue, year, geography, language, and source type. These statistics do not correct bias, but they prevent an unbalanced corpus from appearing neutral.

Security begins with the assumption that retrieved content is untrusted. Prompt injection embedded in papers, web pages, metadata, or repository descriptions must not alter system instructions, tool permissions, or selection policy. Documents are parsed in a data context. Tool calls are generated from trusted control logic and validated against allowlists. The system records which source content influenced a query or action.

Retrieval poisoning is a direct threat. Attackers can publish keyword-optimized documents, duplicate claims across domains, manipulate citations, or create pages designed to occupy the top ranks. Research on poisoning knowledge bases and retrieval corpora shows that RAG systems can be steered through small numbers of malicious documents [52–54]. EPR adds both resilience and vulnerability. Lineage and redundancy controls can reduce a document swarm, but role-conditioned retrieval creates predictable slots an attacker may try to fill.

Portfolio poisoning should therefore be tested explicitly. Adversarial documents can simulate a contradiction, fake an independent replication, or impersonate an authoritative standard. Source authentication, repository trust tiers, version signatures, citation verification, and cross-source corroboration reduce risk. No credibility model should allow polished prose to substitute for verifiable provenance.

Data exfiltration is another risk. Query generation can expose confidential research ideas to external search providers or model APIs. Prior-art searches may reveal invention details before filing. Peer-review material may be confidential. The architecture should support local encoders, private indexes, redacted query plans, controlled external search, and policy-based routing. Logs themselves may contain sensitive ideas and require access control and retention limits.

Copyright and licensing affect both ingestion and output. Indexing rights, storage of full texts, transformation, quotation, and distribution vary by source and jurisdiction. The portfolio can store identifiers and local embeddings while retrieving licensed text at use time. Generated reports should quote only what policy permits and preserve attribution. Scientific utility does not override legal constraints.

Privacy arises in biomedical and social-science evidence. Papers, supplementary data, clinical records, and internal research notes can contain personal or sensitive information. Retrieval indexes can leak rare strings or reveal membership. Access controls must apply before candidate generation, not after selection. A portfolio cannot include evidence the user is not authorized to see. Provenance logs should avoid copying unnecessary sensitive content.

Model governance matters because EPR uses probabilistic components for decomposition, role classification, reranking, and synthesis. Model versions, prompts, thresholds, and calibration sets should be recorded. Upgrades require regression tests on role accuracy, coverage, abstention, bias, and security. A better general language model can worsen a specialized selection policy by becoming more confident in plausible but unsupported role assignments.

Scientific accountability also requires preserving negative results from the system itself. If an EPR selector fails to outperform MMR in a domain, the result should influence deployment. The programme should maintain a capability matrix rather than present one success as universal. Users should know that a portfolio mode is validated for patent component coverage but experimental for biological analogies, for example.

Automated novelty assessment creates special ethical concerns. Labels such as “derivative” or “low novelty” can be socially and professionally damaging. The system should not rank researchers or manuscripts through an opaque novelty score. It should present evidence and conditional comparisons, preserve appeal, and avoid inferring intent. Similarity to prior work can reflect convergent discovery rather than misconduct.

Peer-review automation also requires role clarity. An EPR-assisted reviewer can find missing work and contradictory evidence, but the human reviewer remains responsible for interpretation. Journals should disclose automated assistance according to policy, protect manuscript confidentiality, and prevent retrieved material from becoming an unacknowledged source of review text. Systems should monitor whether they disproportionately challenge work from underrepresented fields because their terminology differs from dominant corpora.

In scientific discovery, portfolio systems can narrow exploration by repeatedly retrieving what the corpus already considers relevant. Explicit analogue and anomaly roles can counter this conservatism, but they can also produce superficial cross-domain associations. Discovery outputs should distinguish literature-supported mechanisms from speculative transfers. Experiments, not retrieval scores, decide whether an analogy is productive.

Auditability requires reproducible corpus snapshots. Web search results change, indexes update, and documents disappear. For important decisions, the system records enough identifiers and hashes to reconstruct the evidence set. When legal or licensing limits prevent storing the source, it records the query, provider, timestamp, and stable identifier. Reproducibility may be approximate for open-web search, and that limitation should be stated.

A portfolio ledger can capture the lifecycle from proposal to decision. It records the task contract, query decomposition, candidate retrieval, exclusions, role assignments, user corrections, selected evidence, synthesis, and final judgement. The ledger need not

be a blockchain; append-only versioning and signatures may be sufficient. The purpose is traceability, not technological spectacle.

Governance should include red-team scenarios. An attacker floods the corpus with duplicated supportive claims. A source contains prompt injection. A famous retracted paper dominates dense retrieval. A confidential invention is sent to an external API. A model labels out-of-domain material as highly novel. A portfolio contains five papers that all reuse one dataset. A contradictory source is excluded because it uses different terminology. Each scenario maps to a test, control, and residual risk.

Human factors matter as much as technical controls. Interfaces that present a polished synthesis first can encourage automation bias. A better design shows the evidence map, uncertainty, and missing roles before the conclusion. Users can inspect a concise narrative, but the system makes it easy to challenge the evidence basis. Confidence visualization should not imply precision unsupported by validation.

The governance objective is to keep the system's influence inspectable and correctable. Evidence selection, reasoning, and decision should remain separable; scope and uncertainty should be visible; source changes should propagate to affected portfolios; and insufficient evidence should lead to an explicit abstention or escalation. These properties provide practical controls for the use of EPR in scientific automation.

Chapter 14. Research Agenda and Technology Development

The research programme begins from a narrow claim: scientific decisions that depend on distributed and contrasting evidence may benefit from retrieval optimized for the collective sufficiency of a bounded portfolio rather than only for independent document relevance. The programme does not assume that a new retrieval model is required. Its first obligation is to determine whether explicit evidence roles, dependence controls, and gap-directed iteration improve downstream work beyond strong RAG and diversification baselines.

The programme should be organized around a shared experimental platform rather than separate demonstrations. The platform ingests scientific papers, patents, standards, reviews, datasets, and internal research artefacts while preserving source structure and provenance. It exposes interchangeable lexical, dense, late-interaction, graph, and citation retrieval channels. It freezes candidate pools for controlled comparison. It supports task-specific role schemas and deterministic portfolio policies. It records every selection decision and exports evidence packages to human and automated reasoners.

The first research question concerns task identification. Not every research query needs a portfolio. The programme must define and detect the regimes in which top-k retrieval is structurally inadequate. Candidate indicators include multiple explicit facets, several claims, requests for comparison or assessment, novelty language, need for counterevidence, multi-hop dependencies, temporal restrictions, and requirements for independent support. A routing benchmark should include direct factual questions, synthesis questions, novelty cases, prior-art cases, review tasks, and discovery prompts. The router is evaluated on both quality and avoided cost.

The second research question concerns the role schemas. Roles should correspond to observable contributions to expert reasoning, not attractive labels invented by system designers. The programme should begin with three target workflows: novelty and idea evaluation, peer-review evidence preparation, and prior-art research. Each workflow is studied through interviews, annotated cases, and think-aloud protocols with domain experts. The output is a compact role schema, annotation guide, and set of difficult boundary examples.

The novelty schema should distinguish direct precedent, component precedent, combination precedent, structural analogue, enabling knowledge, boundary case, independent rediscovery, contradiction, and unfilled gap. The peer-review schema should distinguish supporting evidence, independent replication, failed replication, methodological critique, confounder, alternative explanation, benchmark limitation, population boundary, and missing comparison. The prior-art schema should align evidence roles with claim elements, combinations, terminology variants, patent families, dates, and non-patent literature. The schemas can share infrastructure while retaining task-specific meanings.

The third research question concerns candidate recall. Before optimizing portfolios, the programme must establish whether modern retrieval channels can place the required

evidence in a manageable pool. The platform compares BM25, strong dense encoders, hybrid fusion, ColBERTv2, citation expansion, graph traversal, query decomposition, and agentic search. Recall is measured for complete evidence needs and each role. Failure analyses determine whether missing candidates result from vocabulary, corpus coverage, parsing, temporal filters, or reasoning-intensive relevance.

The fourth question concerns role and dependence prediction. Models are trained or prompted to map candidates to facets and roles, with source-span evidence. Calibration is evaluated across domains. Dependence models identify duplicate versions, patent families, shared datasets, citation derivation, common code, and institutional lineage. The programme should prefer high precision for assertions that affect independence. Uncertain links remain warnings rather than hard exclusions.

The fifth question concerns portfolio selection. The initial selector should be a transparent greedy policy using relevance gates, weighted coverage, redundancy control, source-family constraints, and cost. It is compared with top-k, MMR, xQuAD, set cover, DPP, learned diversification, graph selection, and LLM selection. More sophisticated policies are introduced only when the baseline exposes a repeatable failure. The research contribution may ultimately be a policy, a training objective, a benchmark, or a workflow rather than a new neural architecture.

The sixth question concerns residual-directed retrieval. After selecting an initial set, the system identifies unmet roles and weak facets and generates targeted searches. The programme tests whether this loop recovers decisive evidence that one-shot decomposition misses, whether the generated searches drift from the task, and how to stop efficiently. A controlled comparison fixes total search calls so that iterative retrieval is not rewarded merely for spending more.

The seventh question concerns downstream value. The same reasoner receives evidence from each retrieval condition. Novelty evaluators judge component coverage and the distinctness of the residual contribution. Peer-review agents identify flaws and missing literature. Prior-art analysts assess claim-element coverage. The programme measures not only final accuracy but evidence use, citation correctness, confidence, and time-to-decision. Human experts evaluate whether the portfolio changed or clarified their judgement.

The eighth question concerns portfolio interfaces. A portfolio view differs from a generic ranked-results page by presenting the target decision, role map, facet map, source families, conflicts, gaps, and evidence bundles. Users can replace items, reclassify roles, merge duplicates, and request targeted searches. Interface experiments compare a standard ranked list, an automatically generated synthesis, and the structured portfolio. The primary outcomes are decision quality and correction effort.

The ninth question concerns learning from corrections. Expert changes become versioned training examples. The system learns ranking preferences, role boundaries, and dependence signals without treating every correction as universally applicable. Active learning selects cases where role uncertainty or portfolio disagreement is high. A benchmark split by time evaluates whether the system improves as a research programme accumulates decisions while avoiding leakage from future literature.

The tenth question concerns integration with executable research artefacts. Evidence units should eventually connect to datasets, code, workflows, hypotheses, claims, and verification results. The programme defines identifiers and provenance interfaces that allow a portfolio member to point to an executable check. This integration should remain optional during early retrieval experiments so that the value of EPR is not confounded with a full executable-science platform.

The work can proceed through five experimental stages. The first stage builds the baseline platform and evaluation harness. The second creates expert-annotated task sets and role schemas. The third implements and ablates portfolio selectors. The fourth integrates portfolios with scientific agents and human interfaces. The fifth conducts external, cross-domain validation and operational pilots. These stages are sequential in evidence, not necessarily in software development; interface prototypes can begin early, but deployment claims wait for external results.

Experimental stage	Main question	Required output	Continuation condition
Baseline and instrumentation	Can the relevant evidence reach a controlled candidate pool?	Reproducible hybrid RAG stack, frozen pools, cost and provenance logs	Strong recall and stable baselines on target corpora
Role and benchmark construction	Do experts agree on evidence needs that top-k does not represent?	Annotation guides, hidden test cases, role and dependency labels	Usable agreement and demonstrable distributed-evidence cases
Portfolio selection	Does explicit set construction improve evidence sufficiency?	Greedy and learned selectors, ablations, failure taxonomy	Material gain over reranked RAG, MMR, and LLM selection
Scientific-agent integration	Does the portfolio improve actual novelty, review, or prior-art decisions?	Agent workflows, evidence accounting, human interface	Better downstream quality at equal evidence budget
External validation	Does the effect generalize beyond the designers and one domain?	Independent evaluation, released benchmark artefacts, deployment constraints	Replicated gain in at least two domains and acceptable cost

The first corpus should be scientifically meaningful and operationally manageable. A suitable starting point is an open-access subset in computer science and biomedicine, supplemented by a patent corpus for prior-art tasks. Documents should be versioned and linked through citations, datasets, and code where possible. A bounded, well-audited corpus is preferable for the initial experiments because it supports more interpretable error analysis than an unrestricted scholarly web crawl.

Annotation should use cases rather than isolated passage labels. Experts receive a research idea, manuscript claim, or patent-like description and construct an evidence dossier under a fixed budget. They identify why each item matters and what remains missing. Multiple experts independently create portfolios, then discuss disagreements. The result captures alternative valid sets rather than imposing one perfect ordering.

Candidate-pool annotations can be added after the portfolio task to avoid anchoring experts on system results.

The novelty benchmark should include ideas with a complete precedent, ideas assembled from known components, ideas using old mechanisms under new terminology, genuine-looking but out-of-domain cases, temporally novel ideas that later became common, and cases where experts disagree about the significance of the difference. Evaluation focuses on evidence coverage and calibrated comparison, not a binary claim that the system discovered global novelty.

The peer-review benchmark should contain manuscripts or controlled variants with hidden flaws. These include missing related work, unsupported generalization, benchmark leakage, inappropriate baselines, dependent evidence presented as replication, contradicted claims, and incorrect citations. The retriever is evaluated on whether its portfolio places the decisive external evidence within the reviewer's budget. Review prose is secondary.

The prior-art benchmark should use claim-like descriptions and expert-identified references. It should include terminology drift, distributed claim elements, patent families, non-patent literature, and date constraints. Professional searchers or patent specialists should evaluate whether the portfolio reduces missed material and search effort. The system should not automate legal validity conclusions.

A common technical architecture supports these benchmarks. Document ingestion preserves structure, versions, citations, and rights. Index services provide lexical, dense, and late-interaction search. A task-model service produces editable facets and roles. Candidate services retrieve through role-conditioned queries. An evidence normalizer groups versions and source bundles. A classifier annotates roles, scope, and dependencies. A selector builds portfolios under policy. A residual controller issues bounded follow-up searches. A provenance service records the trace. Downstream agents consume the same portfolio format.

The programme should define a portable Evidence Portfolio Interchange Format. It can be serialized as ordinary structured data, but the conceptual fields matter more than syntax. The format contains the task contract, corpus scope, evidence units, exact source locations, roles, facets, relations to target claims, confidence, dependence links, selection rationale, gaps, and version history. A stable format allows different retrievers, agents, and interfaces to compete over the same task.

The initial product prototype can target novelty and related-work research for AI and software research, where open corpora and expert users are available. A researcher submits an idea or draft contribution. The system returns a portfolio containing direct precedents, component precedents, analogues, contradictions, and a gap report. The user corrects the task model and evidence roles. The corrections feed benchmark construction. The product is positioned as research assistance, not certification.

A second prototype can support peer-review preparation. It analyzes a manuscript's major claims, proposes evidence needs, and builds claim-specific portfolios. Reviewers inspect the evidence and choose which issues to pursue. The prototype records whether the portfolio exposed a flaw or missing citation that ordinary search or the reviewer initially

missed. Confidentiality and source-access policy must be part of the design from the beginning.

A third prototype can support internal idea evaluation in research organizations. It builds balanced dossiers for proposed projects, including precedents, technical dependencies, failure evidence, alternative approaches, and validation methods. This use case is less constrained by formal publication workflows and can provide rapid feedback on interface and portfolio utility. It should not become an opaque project-ranking system.

The programme's scientific outputs should extend beyond a trained model. A task taxonomy, annotation methodology, benchmark corpus, evidence-role schema, portfolio interchange format, evaluation harness, and documented negative results may be more durable than any single selector. The software should support reproducible baselines and external contributions, and its modular architecture should permit new retrieval models to be evaluated without changing the research question.

The principal risks are substantial. The role schema may be too subjective. LLM rerankers may infer equivalent portfolios without explicit portfolio machinery. Expert portfolios may be expensive and non-unique. Corpus incompleteness may dominate novelty errors. Dependence detection may remain unreliable. Portfolio interfaces may increase cognitive load. Gains may diminish with larger context windows or fail to transfer across domains. These risks justify staged experiments, explicit baselines, and predeclared continuation criteria.

The stop conditions are equally important. The EPR-specific direction should be reduced or closed if a strong hybrid and late-interaction RAG with query decomposition and MMR matches downstream performance; if role annotations have unusably low agreement; if selector gains disappear outside the development domain; if experts do not find the portfolio more useful; or if cost increases are disproportionate. In that outcome, the surviving contribution may be an evaluation framework and a better diversified RAG workflow.

The positive outcome is also clear. EPR merits continued development if explicit portfolio construction repeatedly improves scientific decisions at the same context budget, if the gain is attributable to role and dependence structure rather than general model strength, if the system knows when it lacks evidence, and if users can inspect and correct the result. That outcome would justify dedicated portfolio models, learned policies, and integration across research agents.

The programme is ambitious in application and conservative in inference. It aims to improve novelty research, prior-art search, peer review, discovery, and idea evaluation while treating each architectural claim as provisional. Its central deliverable is an evidence infrastructure that gives humans and agents a better basis for making judgements about what is new, credible, or worth testing, with the supporting sources and unresolved limitations kept visible.

Chapter 15. Conclusions: Retrieval as Evidence Selection for Scientific Decisions

RAG changed the practical relationship between language models and external knowledge. It made retrieval part of generation, enabled source-grounded answers, and created an engineering discipline around corpora, indexes, rerankers, context assembly, citations, and evaluation. By 2026, RAG is not one pipeline but a family that includes hybrid search, late interaction, adaptive retrieval, hierarchical indexes, graph methods, agentic search, multimodal evidence, long-context hybrids, and verification loops. Any new proposal must be compared with this mature and rapidly moving baseline.

The limitations of RAG are also well characterized. Relevant evidence can be absent from the corpus, altered by parsing, split by segmentation, missed by representation, excluded by early ranking, diluted by context, ignored by the generator, cited incorrectly, or manipulated by adversarial content. Advanced architectures reduce some of these risks while introducing different costs and failure modes. Appropriate responses include instrumentation, task-specific design, competitive baselines, and evaluation across the full information path.

Evidence Portfolio Retrieval begins from one remaining mismatch. Scientific decisions often require a set whose members play complementary roles. The closest documents are not always the best dossier. A novelty assessment needs complete and component precedents, combinations, analogues, and boundary cases. A peer review needs support, contradiction, replication, methodological critique, and scope. A discovery workflow needs mechanisms, anomalies, confounders, interventions, and discriminating measurements. Individual relevance remains necessary, but collective sufficiency becomes the target.

EPR is therefore not an alternative search engine. It is a control layer over candidate retrieval and an output contract for reasoning-ready evidence. It decomposes a task into evidence needs, retrieves broadly through conventional channels, classifies candidates by facets and roles, models versions and dependencies, selects a bounded complementary set, and reports unresolved gaps. It can use MMR, set cover, graphs, LLMs, or learned policies. Its identity lies in the task objective and accountability structure, not in one algorithm.

This framing separates two issues that are often conflated. Diversity does not by itself establish epistemic quality, because documents can be mutually different and still irrelevant to the decision. EPR therefore retains relevance gates and task-specific roles. Likewise, absence from a bounded corpus does not establish novelty. EPR can report that a relation remains uncovered after a documented search protocol, but it cannot establish that no such relation exists outside the searched material.

The engineering opportunity is substantial because context remains scarce even when windows grow. Attention, latency, cost, and human verification impose budgets. A

compact evidence package can be reused across agents, reviewed by humans, updated when sources change, and connected to executable artefacts. In a research environment, the portfolio can become the shared object linking novelty analysis, peer review, compliance, planning, and discovery.

The comparative advantage of EPR remains to be established. Modern RAG already includes query decomposition, multi-hop retrieval, late interaction, graph selection, diversification, and agentic search. EPR must therefore be compared with combinations of these methods under equal evidence, cost, and latency budgets. The evaluation must determine whether explicit roles, source-independence modelling, contradiction search, and residual iteration provide value beyond generic reranking or LLM-based selection.

Negative results should lead to architectural simplification. If modern RAG combined with MMR or an LLM selector produces equivalent portfolios, the task taxonomy and evaluation framework may remain useful while the distinct selection layer is removed. If explicit portfolio construction improves only novelty and prior-art workflows, it should be limited to those applications. The appropriate scope should be determined by the observed capability boundary.

The most promising near-term product is an evidence dossier builder. It accepts an idea, manuscript claim, or invention description and returns a small set of direct precedents, component evidence, analogues, contradictions, dependencies, and gaps, all linked to exact sources. It does not certify novelty or write the final review. It prepares a more balanced and inspectable substrate for those decisions.

The deeper research direction is retrieval optimized for downstream epistemic work. This moves evaluation from “Did the system rank relevant documents?” to “Could the decision maker reach a better, better-supported conclusion using the selected evidence?” It requires new benchmarks, expert annotations, set-level metrics, source-dependency models, and careful human studies. It also provides a principled interface between retrieval systems and the emerging generation of research agents.

The long-term vision of executable science strengthens this need. As claims, methods, data, code, provenance, and inference become increasingly machine-readable, retrieval can assemble not only passages but verifiable scientific objects. A portfolio may contain a claim, supporting studies, executable reproductions, contradictory results, and the experiment that would resolve the disagreement. The system does not replace science. It organizes the evidence so that scientific reasoning can become more explicit, testable, and reusable.

RAG remains the principal foundation for retrieval-grounded language systems. EPR is a distinct, testable proposal about how evidence should be selected when the downstream objective is a scientific judgement rather than a direct answer. The proposal is relevant because distributed and contrasting evidence is common in novelty assessment, prior-art research, peer review, and scientific discovery. Its future should be determined by comparative experiments that measure whether evidence portfolios improve these decisions under controlled budgets and with adequate provenance, calibration, and external validation.

Glossary

Term	Engineering meaning in this book
Approximate nearest-neighbour search	An index that finds likely close vectors without comparing the query with every stored vector. It trades a small amount of recall for speed and scale.
BM25	A sparse lexical ranking function that rewards query-term matches while accounting for term rarity and document length. It remains a strong baseline for exact terminology.
Candidate pool	The broader set of potentially useful evidence produced before expensive reranking or portfolio selection.
Chunk	An indexed segment of a source. Its boundary affects whether evidence and its conditions remain together.
ColBERT	A late-interaction retrieval architecture that preserves multiple token-level representations and matches query tokens against document-token representations.
ColBERTv2	A more storage-efficient and robust development of ColBERT using residual compression and denoised supervision.
Complete-evidence recall	Whether the candidate pool contains all evidence units required to answer or assess a task, not merely one relevant item.
Context budget	The maximum evidence, usually measured in tokens or passages, supplied to the downstream model or human.
Cross-encoder	A reranking model that jointly processes a query and candidate text, often improving precision at higher computational cost.
Dense retrieval	Retrieval based on learned continuous vector representations rather than exact term overlap.
Dependency or lineage	A relation indicating that apparently separate sources may derive from the same data, experiment, patent family, review, or intellectual origin.
Diversified retrieval	Selection that balances relevance with coverage of different subtopics, intents, or result types.

Term	Engineering meaning in this book
Evidence bundle	A coherent package of source passages, tables, figures, or metadata needed to interpret one evidential contribution.
Evidence Portfolio Retrieval	Selection of a bounded, complementary, role-aware set of evidence optimized for a downstream decision rather than only for individual document relevance.
Evidence role	The function an evidence unit plays relative to a task, such as direct precedent, contradiction, replication, methodological critique, or boundary case.
Facet	A component or aspect of the target question, claim, or idea that evidence can cover.
Faithfulness	The degree to which a generated answer is supported by the supplied evidence rather than by unsupported model generation.
GraphRAG	A family of systems that organizes or retrieves evidence through graph structures, often using entities, relations, communities, or paths.
Hybrid retrieval	Combination of lexical and semantic retrieval channels, often followed by fusion and reranking.
Late interaction	Query and document retain multiple representations and are compared through fine-grained interactions at retrieval time rather than being reduced immediately to one vector.
Maximal Marginal Relevance	A greedy method that balances query relevance against similarity to already selected items. It is a central diversification baseline for EPR.
Multi-hop retrieval	Retrieval in several dependent steps, where evidence found in one step helps formulate or constrain the next.
nDCG	Normalized Discounted Cumulative Gain, a ranking metric that rewards highly relevant items appearing early and supports graded relevance judgements.
Novelty residual	The explicitly described part of an idea not covered by the current evidence portfolio. It is a search gap, not proof of scientific novelty.
Out of distribution	Input that falls outside the domains or patterns represented by the system's corpus or training data. Distance from the KB can indicate this condition rather than novelty.

Term	Engineering meaning in this book
Portfolio sufficiency	Whether the selected evidence is adequate for the target judgement under the declared corpus and budget.
Provenance	The source, version, location, retrieval trace, and transformation history associated with evidence and derived outputs.
Query decomposition	Transformation of a complex request into facets, subquestions, relations, constraints, or evidence roles.
RAG	Retrieval-Augmented Generation, a family of systems that conditions model output on evidence retrieved from external stores.
Reranker	A model or algorithm that reorders candidates produced by a faster first-stage retriever.
Residual-directed retrieval	Additional search driven by evidence roles or facets still missing after the current portfolio has been inspected.
Role coverage	The fraction of required evidence functions represented credibly in the portfolio.
Source-span support	Exact text, table, figure, or data location that supports an extracted claim or role assignment.
Sparse retrieval	Retrieval over explicit terms or learned sparse weights, enabling exact matching and inverted-index execution.
Top-k retrieval	Selection of the k candidates with the highest individual ranking scores.
Vector database	Storage and search infrastructure optimized for embedding vectors and associated metadata.

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